



Overview of

The Indian Buffalo

Meat Value Chain



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Message

The livestock wealth of India is one of the highest in world. India is the world's largest exporter of buffalo meat and accounts for 58 per cent of the world's buffalo population. Buffalo in India contributes about 30% of total meat production in the country. However despite this potential and growth, the sector is not well integrated. The present system of production and marketing of buffalo meat for domestic and export market is endowed with multifarious challenges and needs corrective measures at various levels. There are multiple stakeholders along the buffalo meat value chain, but most of them operate in isolation and lack information at various levels along the chain. Therefore, it is difficult to assess how problems identified at a single stage of the value chain compare to and interact with problems in the rest of the value chain.

Having realised this fact, FICCI's Agriculture Division has undertaken a comprehensive survey in major meat producing states of the country interviewing major stakeholders of the buffalo meat value chain and developed an in-house report on Indian buffalo meat value chain. The study not only identified the problems in different parts of the value chain but also elicited suggestions from stakeholders on what can be done further to strengthen the sector.

Time has come to seriously consider and ensure hygienic meat production in the country. While there is tremendous potential of growth of buffalo meat production in the country, steps should be taken to develop improved systems of meat production, handling and marketing to give a fillip to meat exports besides also meeting local demand. Proper utilisation of animal by-products and animal wastes could be extremely vital and useful from economic point of view.

FICCI has always thrived in providing thought leadership. This study is an initiative on part of FICCI to bring out an integrated analysis of the value chain of buffalo meat production for domestic as well as export market. I hope that the information given in the report will be useful to stimulate new ideas as well as contribute to the development of more innovative value chain of buffalo meat in India.

A. Didar Singh



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1

Preface

01 Preface



The livestock sector is an important component of India's economy in terms of income, employment and foreign exchange earnings. An analysis of the composition of agricultural trade over the last decade shows that traditional agricultural exports of India have been overtaken by new and more dynamic sectors, such as buffalo meat exports.

Agriculture continues to be the primary industry in India which directly or indirectly supports more than 50 percent of the Indian population. However, agriculture in India has always been heavily dependent on the monsoons and has hence been an inherently risky activity. In this case, diversification by increasing the share of income from allied sectors becomes important.

Besides directly benefiting the rural economy, the buffalo meat industry also provides raw material for allied sectors. For instance, high-quality hides for the leather industry, processed offals for pharmaceutical and pet food applications, and finally rendering of inedible by-products provides poultry feed, special lubricants and other products.

Driven by the structural changes in agriculture and food consumption patterns, the importance of buffalo meat sector has been undergoing a steady transformation. In recent years the buffalo meat industry for export has attracted heavy investment as we see major buffalo meat exporters expanding their processing capacities to meet demand in new worldwide markets. The Government of India is also taking steps to provide thrust to the buffalo meat sector by introducing schemes on salvaging and rearing of buffalo male calves and modernization of abattoirs.

The sector has significant potential to grow further. The expanding domestic market as well as export markets for buffalo meat is creating a window of opportunity for all the actors in the value chain of buffalo meat production. This report provides an overview of the value chain of two production models: municipal slaughterhouse system for domestic consumption and large-scale production for export.



2

Executive Summary

02 Executive Summary

Meat production in India including poultry meat is estimated at 4.9 million tonnes (2010-11), standing eighth in rank in the world's meat production. Buffalo contributes about 30% of the total meat production in the country.

The buffalo meat sector therefore is one of the important sectors in Indian economy. However, with changing world food production and consumption patterns, there is a need to develop a strategy for the future by analysing different segments of the value chain of buffalo meat production.

Existing studies of the sector have examined only a single stage of buffalo meat production. Therefore, it is difficult to assess how problems identified at a single stage of the value chain compare to and interact with problems in the rest of the value chain. Against this background, FICCI carried out a study to understand the value chain of buffalo meat production for the domestic as well as the international markets. The report is divided into three sections. The first portion describes the value chain of buffalo meat production and processing for the domestic market: stakeholders involved in the value chain, their roles and relationships with other actors, movement of products and costs involved at each stage. The subsequent section describes the same pattern in the buffalo meat value chain for export. The final section focuses on emerging trends in demand for buffalo meat in the domestic and export markets and India's production capacity.

Indian buffalo meat value chain for domestic market

The domestic value chain for buffalo meat production in India consists of a network of stakeholders, who work within specific stages from farm to plate. The chain integrates livestock producers, traders, butchers, wholesale meat dealers and retailers. This section sheds light on the roles and responsibilities of various actors as the product moves from farm to consumers in domestic value chain and also various stages and the economics involved in each stage.

India's livestock sector has continuously provided structural support to the rural economy as an important vocation for rural India, next only to crop raising. In view of this significance of buffalo in the domestic market, it is imperative for policy makers to devise policies and frameworks that would ensure sustained momentum in this sector. Therefore, the next phase of active reforms in the buffalo meat value chain for the domestic market should consider (a) modernisation of existing abattoirs, (b) strengthening of livestock markets, (c) utilisation of by-products (d) an efficient disease diagnostic and monitoring system, and (e) developing a state-wise market intelligence system for the livestock sector, which can assist in realistic planning at the micro and macro levels.

Indian buffalo meat value chain for export market

The buffalo meat value chain for export has different stakeholders working at several stages of production. Farmers raise the buffalo till they become uneconomical and thereafter sell them to traders. Traders play the role of aggregators in the value chain and sell the buffalo to exporters for further processing and sale in export markets. The chapter sheds light on costs involved in undertaking different operations in modern integrated abattoirs and increases in costs of various operations from 2010 to 2012. An increase of 27% has been observed in the cost of buffalo from 2010 to 2012 whereas the cost of electricity and labour has increased by 25%, transport costs by 20%, packaging costs by 13% during the same period.

The upward trend in the Indian buffalo meat sector is clearly reflected in the export figures. In FY 2012-13 buffalo meat valued at ₹ 17,400.59 crores (USD 2.9 billion) was exported. However, as we enter the next phase of development in this sector, which will see the more active participation of India in the international buffalo meat markets there is need to strengthen the ongoing reforms by active participation of all the stakeholders.

India's production capacity

High mortality of buffalo calves was revealed in the study conducted by Central Leather Research Institute (CLRI), Chennai in 2010. These calves could otherwise be salvaged for meat production and recovery of hides, thereby improving the economic condition of farmers and providing quality meat for the export market and domestic consumption. As per the available data, 14 million male calves perish annually. If 70% can be salvaged, reared and processed @ 175 kg per head – including edible offals – when reared they can yield 1.72 million MTs meat per annum. Therefore, sustainable livestock development requires integrated efforts for implementation of a male buffalo calf rearing program for the long-term benefit of the farmers, country and the trade.

Future prospects of Indian buffalo meat sector

Indian buffalo meat is one of the largest agricultural exports (by value) in the country, after guar, rice and seafood. In coming years the sector is expected to grow at a CAGR of 15%. Active participation of the private sector along with government schemes on salvaging and rearing of male buffalo calves is expected to augment the availability of livestock in coming years. Unlike in previous years, livestock farmers will consider raising male buffalo calves as a remunerative activity. The focused implementation of the Ministry of Food Processing, Government of India scheme for “Setting up and Modernization of Abattoirs” will certainly provide impetus to the buffalo meat sector in domestic market. Animal welfare has become an integral part of the Indian livestock production system with established rules on cruelty to animals and transportation. Buffalo slaughtered in India are a by-product of the dairy industry and the average carcass yield per animal varies. APEDA anticipates a pan India average of 150 kg boneless meat and edible offal per buffalo in coming years. Advance preparatory actions are already in place for complete eradication of FMD from the country. Further, amendments to the Meat Export Policy would certainly help the buffalo meat industry gain its rightful position in target export markets.

The report aims to provide data on the main marketing channels various cost components and an integrated outcome of how the value chain of buffalo meat production behaves for the domestic as well as the international market.



3

Research Methodology and Approach

03 Research Methodology and Approach

A. Research objectives:

1. To study the value chain and analyse the cost of production of domestically consumed Indian buffalo meat (carabeef).
2. To study the value chain and analyse the cost of production of export-oriented Indian buffalo meat (carabeef).
3. To generate information on areas for intervention in the value chain of buffalo meat production.
4. To review demand for buffalo meat in the domestic and export markets and India's production capacity.

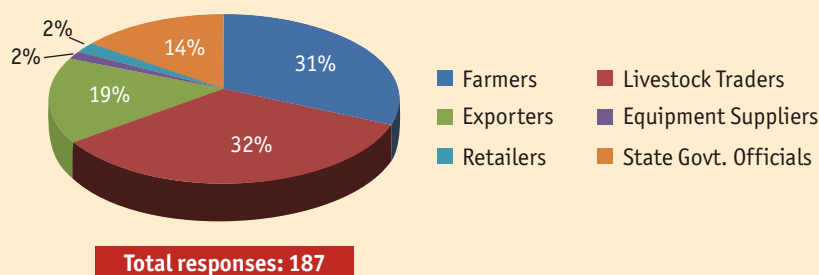
B. Approach and methodology:

1. The study on analysing various points of buffalo meat value chain involved meticulous collection and analysis of secondary and primary data. The secondary data used both the quantitative and qualitative data. Primary data collection was done with the aid of a printed questionnaire designed for different categories of respondents. The tools used for primary data collection included personal interviews and group discussions.
2. The draft questionnaire was initially field tested with a few respondents. Based on the feedback the questionnaire was improved and finalised.
3. Questionnaire design was aimed at eliciting accurate answers from the respondents which can further be used in making decisions. Various types of questions were used while structuring the questionnaire for (a) open ended questions, (b) close ended questions, (c) probing questions, (d) dichotomous questions, and (e) multiple choice questions.

C. Sample composition and sample size:

For undertaking the survey a questionnaire was circulated and interviews were organised with major stakeholders of the buffalo meat value chain such as livestock farmers, traders, retailers, equipment suppliers, central and state government officials during the months of February through June 2013. In total 187 responses from the buffalo meat sector were received. The detail of sample composition and sample size (in percentage) is provided in the below graph.

Sample composition and sample size



State selection criteria: The sample states were selected taking the following parameters into account:

- (a) States with high production of buffalo meat (carabeef),
- (b) States with the largest number of export-oriented production units,
- (c) States with maximum population of Muslim communities, as consumption of buffalo meat in India is limited mainly to Muslims and certain other minority communities, and
- (d) States which are main procurement centres for live buffalo.

State Selection Criteria

State	Carabeef produced, % of total, 2011-12	Muslim population, (% of total muslims in India)	Exporter units, approx number	Remarks (Reasons for short listing)
1 Uttar Pradesh	44	22	22	Production, consumption and export potential
2 Andhra Pradesh	12	5	3	Potential for production, high domestic demand
3 Maharashtra	10	7	2	Production, consumption and export potential
4 Kerala	10	6	1	Potential for production, high domestic demand
5 Bihar	5	10	0	Procurement centre for exporters
6 Delhi*	4	1		Export hub
7 West Bengal	2	15	0	High domestic demand
8 Rajasthan	2	3	2	Procurement centre for exporters
9 Karnataka	1	5	0	High domestic demand
10 Punjab	10	0.2	4	Export potential

Source: APEDA, Census of India 2011, Basic Animal Husbandry Statistics, Department of Animal Husbandry, Dairying and Fisheries (DADF), Ministry of Agriculture, Government of India.

* Meat produced data of 2010-11 is considered

** Note: by short listing these states 70% of buffalo meat production, Muslim-populated areas, and export units are covered.

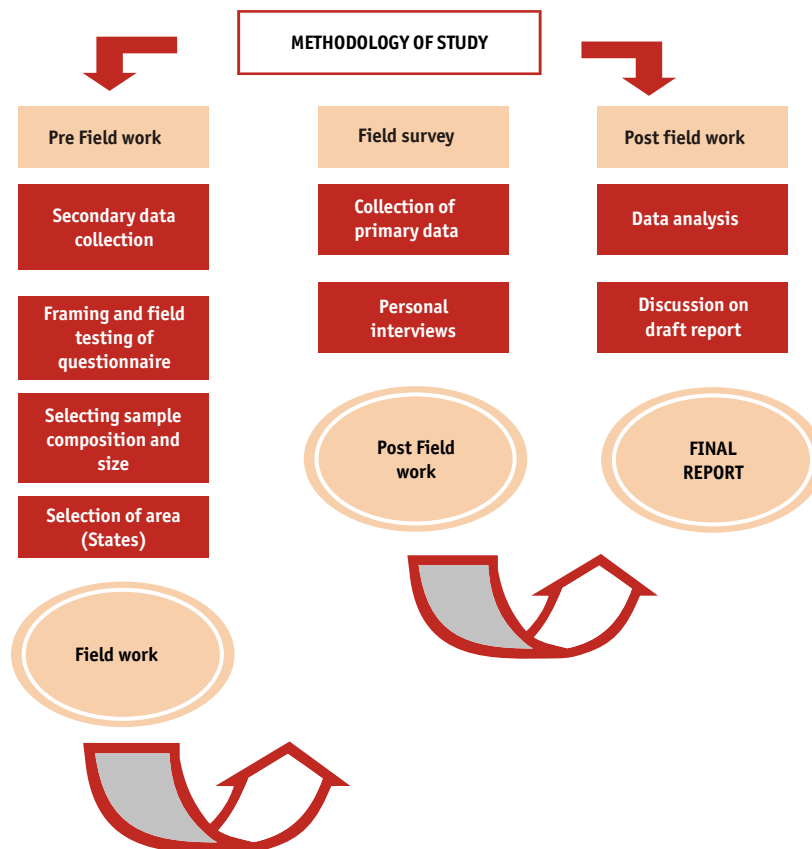
U.P has maximum export units which show acceptability of this industry in that State. Livestock from other states also move into UP for processing purpose.

D. Sampling technique:

1. Stratified random sampling technique was adapted to survey exporters. For this, the list of approved Indian abattoirs-cum-meat processing plants was categorised into three strata according to the slaughtering capacity of buffaloes per day:
 - (a) Class I: 200-400 head/day,
 - (b) Class II: 500-700 head/day,
 - (c) Class III: > 700 head/day.
2. Snowball sampling method was used to survey livestock farmers, traders, equipment suppliers and retailers.

E. Data collection and report writing:

The data collected through questionnaires and personal interviews were analysed. Various interpretations were made based on the outcome of field data. A detailed report was prepared, which yielded some major observations in the value chain of buffalo meat production. The survey was undertaken from February through June 2013.





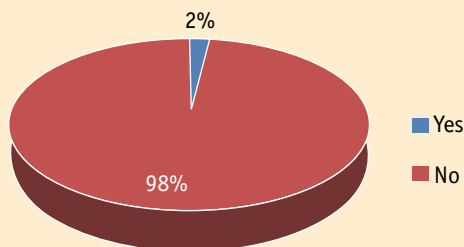
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Stakeholder Analysis

04 Stakeholder Analysis

(1) High mortality of male buffalo calves have been noticed in different parts of the country. These calves could otherwise be salvaged for providing quality meat. Keeping this factor in view, Department of Animal Husbandry, Dairying and Fisheries, Ministry of Agriculture, Government of India has introduced a scheme for "Salvaging and Rearing of Male Buffalo Calves". Livestock farmers were questioned whether they are aware of such scheme on salvaging and rearing of male buffalo calves. Evidence from the survey suggests that only 2% of the surveyed farmers were aware of this scheme.

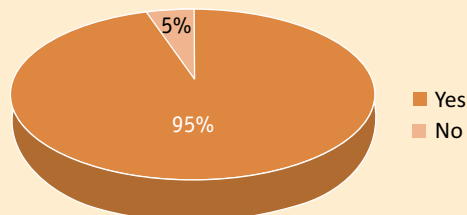
Livestock farmers were asked, "Are you aware of Government scheme of salvaging and rearing of male buffalo calves?"



Small and marginal farmers play a major role in buffalo husbandry presently. This signifies that growth in demand of buffalo meat necessitates an efficient flow of information, services and support to livestock farmers for better decision making. The Government scheme for "Salvaging and Rearing of Male Buffalo Calves" should be implemented expeditiously with active participation of animal husbandry departments of state governments. This will help shape the future growth in animal production in India, as well as emerging trade and investment opportunities.

(2) The Government of India is conscious of the quality issue in meat and is providing incentives to various state governments to ensure scientific and hygienic slaughtering of the animals and supply of quality meat and meat products. The name of the scheme being implemented by the Ministry of Food Processing in this regard is 'Setting up/Modernisation of Abattoirs'. Under this scheme, grant-in-aid assistance of 50% in general areas and 75% in difficult areas, for cost of plant and machinery and technical civil work, up to ₹15 crore for each project, is admissible.

Do you think that scheme for "Modernization of Abattoirs" needs a business approach and wider dissemination at ground level?



Respondents (State government officials and officials working at municipal slaughterhouses) were asked to comment on the functioning of such schemes. As the pie graph above shows, a majority of the respondents said that the impact of such schemes presently is not noticeable. First, there is a need to disseminate information about such schemes at ground level. Second, there is a need to build these schemes with a businesslike approach and high priority should be accorded by state governments to modernization of municipal slaughterhouses.

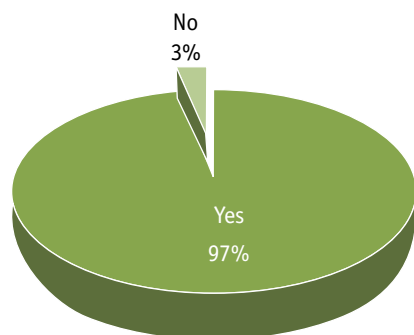
(3) During the survey, the team also realized that there is lack of information on the contribution of each state to the total buffalo meat exports of the country. State Governments should create reliable databases and market intelligence systems for their respective states that can assist in realistic planning at the micro and macro levels.

(4) Exporters were asked about the costliest operation in buffalo meat production besides the cost of live animals. The choice of options given to exporters included (a) labour costs, (b) packaging costs, (c) electricity and diesel costs, (e) service costs and (d) transportation costs.

The answer received from a majority of respondents was, “the cost incurred in transportation from export unit to shipping port” (particularly from North India to port of Western India).

Buffalo meat for the export market is transported to the shipping port in refrigerated containers with a temperature of less than -18 degree Celsius. Container capacity is 28 tons of meat.

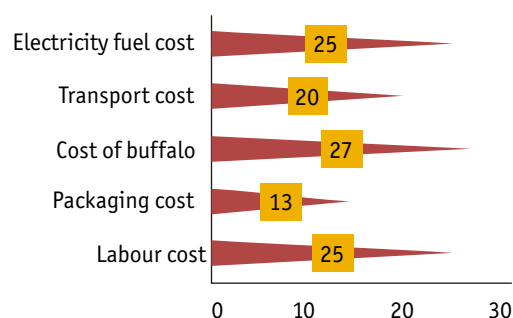
Do you think transport from factory gate to shipping port is the costliest operation in buffalo meat supply chain?



As per the survey results approximately 38% of the expense is incurred in transporting packaged buffalo meat from the factory gate to the shipping port in refrigerated containers.

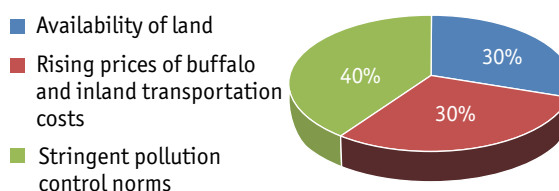
(5) Further, when asked about the extent of increase in costs of major operations in buffalo meat production for export, some major trends emerged.

Percentage increase in cost of various operations from 2010-2012



Exporters reported that the overall cost of production of buffalo meat has increased in the last three years. The percentage increase in the costs of various operations for production of buffalo meat for export is shown above.

What is the major bottleneck in supply chain of buffalo meat for export market?



(6) Exporters of buffalo meat were asked to comment on various bottlenecks in production of buffalo meat for export. The choice of options given to exporters included (a) availability of land, (b) rising prices, (of buffalo and inland transportation costs), and (c) stringent pollution control norms.

Approximately 40% of the respondents raised concerns on zero discharge policy and stringent pollution control norms. While 30% mentions the availability of land and the rest, 30% stresses rising prices of buffalo and inland transportation costs.

According to the zero discharge policy, even after maintaining the BOD (biological oxygen demand) level of less than 30mg/l and COD (chemical oxygen demand) of less than 250 mg/l, and TSS (100 mg/l), oil and grease (10 mg/l) meat packers could not discharge any effluent outside the export unit. On average 600-1000 litres water/day is used for buffaloes in a slaughter house. Hence, it becomes difficult to utilise such a huge quantity of water inside the factory territory. It has been suggested that once the water is treated according to norms of CPCB it should be allowed to drain outside the factory land.

Land being a vital component for establishment of new abattoirs, it becomes a major concern when (a) sufficient land is not available for construction of an abattoir (which includes a modern slaughter hall, lairage, ETP plant, etc.), (b) there is difficulty in identification of a suitable site outside the densely populated areas or away from residential areas, or (c) complex norms for land conversion exist.

Approximately 30% of the respondents showed concern over rising prices of buffalo and inland transportation costs, which increase the overall cost of the value chain.



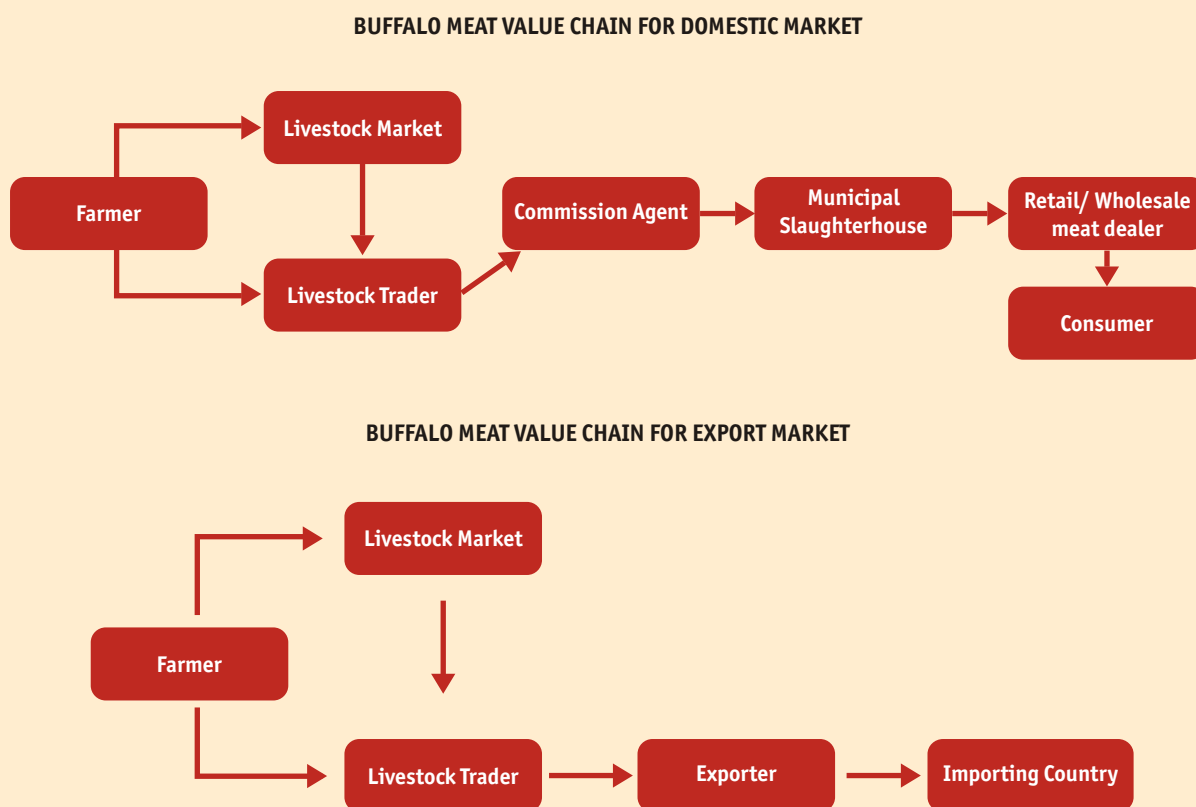
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Indian Buffalo Meat Value Chain for Domestic and Export Market

Indian Buffalo Meat Value Chain for Domestic and Export Market

Structure of buffalo meat sector: Buffalo meat production can be classified into two major value chains: (a) large-scale production for export, and (b) municipal slaughterhouse system for domestic consumption. The third prevalent model is informal smallholder “backyard” slaughtering in remote areas. Backyard slaughter is characterized by low inputs and slaughter of animals for one's own consumption or for sale of surpluses to local markets. Municipal slaughterhouses are State Government bodies that provide services for buffalo meat production for domestic consumption. Production for export consists of companies controlling the initial stage of animal purchase from livestock traders to marketing deboned meat in the international market.

Figure 1. Buffalo meat value chain for domestic and export market



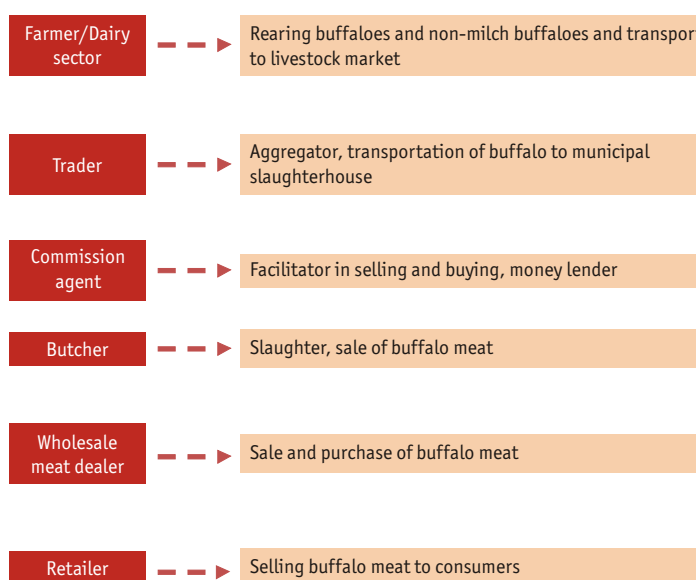
SECTION 5.A

BUFFALO MEAT VALUE CHAIN FOR DOMESTIC MARKET

SECTION 5.A.1 ROLES OF DIFFERENT ACTORS

The domestic value chain for buffalo meat production in India consists of a complex network of stakeholders, who work within specific stages from farm to plate. The chain integrates livestock producers, traders, butchers, wholesale meat dealers and retailers. The livestock (buffalo) are moved from farmers to traders, then to butchers who slaughter animals and convert them into marketable products. The products finally reach consumers through wholesale meat dealers and retailers.

Figure 2. Roles of different actors in Domestic value chain of buffalo meat production

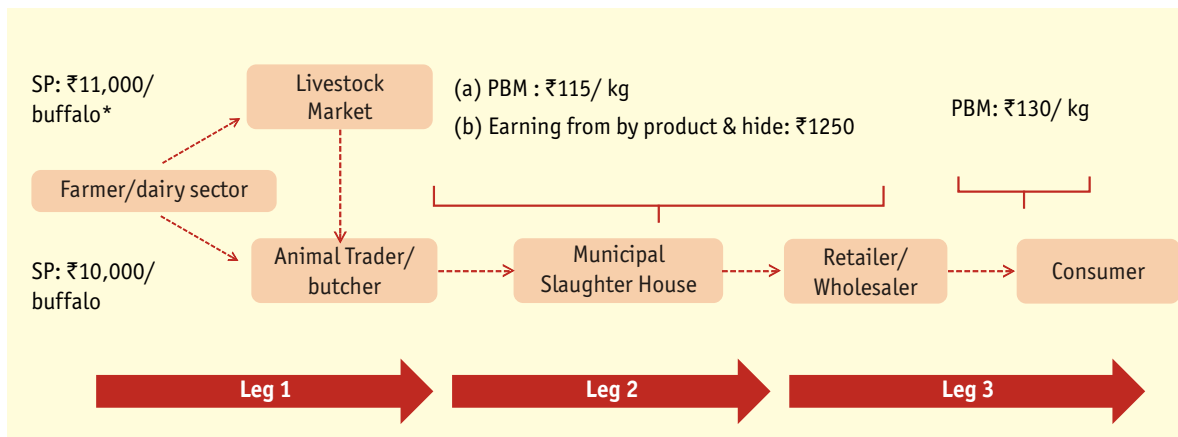


SECTION 5.A.2

STAGES IN BUFFALO MEAT PRODUCTION IN DOMESTIC SECTOR

Value chain: Buffalo meat produced in India is primarily raised by dairy farmers. Most farmers sell their animals to traders. Livestock traders don't own slaughter facilities but rather use the services of butchers at municipal slaughterhouses. The buffalo meat is then collected by wholesale meat dealers or in some instances directly by retailers. This chapter sheds light on the roles and responsibilities of various actors as the product moves from farm to consumers and also various stages in the value chain and the economics involved in each stage.

Figure 3. Stages in buffalo meat production for domestic market and cost involved in each stage



PBM: Price of Buffalo meat, CY: Carcass yield (Carcass yield of 200 kg buffalo is approx 110 kg)
SP: Sales price, *The estimated weight of buffalo in this figure is 200 kg

Leg1: Farmers rear buffalo till they reach an unproductive age and thereafter sell them either to traders or at the livestock market

Role of farmers as primary producers: India has approximately 15% of the world's cattle (genus *Bos* spp.) inventory and nearly 58% (estimated 113 million, FAO 2011) of the world's water buffalo (*Bubalus* spp.) inventory. These animals are generally reared by subsistence farmers with small land holdings, each rearing 2 to 3 buffalo. Though farmers play a crucial role in rearing buffalo, but ironically, raising buffalo, especially for meat production, is not a well planned activity. Buffalo are not raised on dedicated farms for meat production. Feeding consists of dry roughage supplemented with seasonal green fodder and concentrates.

Table 1. Farmgate price of live buffalo for sale in the domestic market

	Liveweight of buffalo, kilograms	Price at livestock market, per head		Trader prices, per head	
		₹	USD	₹	USD
1	200	11,000	183	10,000	166
2	150	9,000	150	8,000	133

Source: FICCI primary survey

*Throughout entire report 1 USD=60 ₹ (as of August 13, 2013)

Leg2: Traders transport the purchased animal to a municipal slaughterhouse for slaughtering.

Role of traders as aggregators and facilitators: The value chain of buffalo meat production for the domestic market is very informal. Traders play a very active role in this value chain as intermediary aggregators. Farmers (primary producers) have limited market access and are isolated from major consumers due to logistical and transport costs. Farmers often avoid these costs directly by selling to intermediary aggregators. Traders buy animals from various farmers and pool them for further marketing or haul purchased animals to municipal slaughterhouses. During transit of the animals from livestock markets to slaughterhouses health certificates from state veterinarians are also obtained.

Different types of traders (middlemen) involved in the value chain are:

- a) **Petty traders** are the traders involved in occasional (irregular) marketing of animals; they transact a few animals on any given market day.
- b) **Commission agents** facilitate animal transactions between sellers (farmers/livestock traders) and buyers (butchers) and charge a commission on every sale.

Butchers play an integral role in the value chain by performing slaughtering operations.

The word butcher denotes the grassroots-level worker performing slaughter and dressing operations of food animals in slaughterhouses. The scope is further extended to include workers who handle carcasses and meat in meat processing establishments and in retail meat stalls. Non-butchers are workers engaged in the meat industry in jobs other than slaughter and dressing operations, such as animal attendants, sanitation staff, by-product handlers, meat packaging staff and support staff working in effluent treatment plants (ETP), cold rooms, and machinery units. Butchers play an integral role in the value chain of buffalo meat production both for domestic as well as export market by undertaking slaughtering operations.

Slaughter of animals is a state subject in India. Slaughterhouses are managed by local bodies such as corporations, municipalities and panchayats. For slaughtering animals, a trader takes the animal to municipal slaughterhouses, wherein a slaughtering fee of a specified amount is to be paid by the trader.

Leg 3: Retailers/wholesale meat dealers purchase buffalo meat (carabeef) from municipal slaughterhouses and sell to consumers.

- a) **Wholesale meat dealers** collect meat from municipal slaughterhouses and resell to retailers in the locality.
- b) **Retailers are both** buyers and sellers. However, in few instances retailers perform the function of slaughtering and wholesaling also.

From slaughterhouses, buffalo meat is purchased by wholesale meat dealers or retailers. Wholesale meat dealers and retailers have to obtain licenses from municipal corporations for buying and selling on the premises of a slaughterhouse. The Indian buffalo meat industry for domestic consumption is based on production of fresh meat which is processed and sold daily. Retail outlets generally are not equipped with air conditioners, refrigerators or deep freezers.

Buffalo meat and edible offal are transported to retail outlets, where they are sold at retail to consumers. The carcasses are generally packed in polythene bags/sheets and transported by various modes such as light delivery trucks ('Tempos'), auto rickshaws, and motorcycles. Unlike mutton stalls, which sell sheep meat, goat meat and dressed chicken, buffalo meat stalls sell only buffalo meat and offal. Buffalo meat is sold at domestic retail market in both boneless and bone-in form.

SECTION 5. A. 3

COSTS INVOLVED AT DIFFERENT STAGES OF DOMESTIC VALUE CHAIN

(1) Cost incurred by farmers:

As soon as dairy buffalo become unprofitable, farmers cull them for meat production. Farmers have the option of selling either at weekly livestock markets or directly to traders. The comparative price received by farmers by selling buffalo at livestock markets and directly to traders is shown in Table 1. An estimated profit and loss sheet of a dairy farmer is given in Table 2. The costs described in the table are based on a buffalo yielding 10 litres of milk per day on average.

Table 2. Expenses incurred by farmers in raising buffalo

Particular	Amount (₹)	Amount (USD)
Cost of productive buffalo of age 3 years	50,000	833
Farmer rears buffalo for approx 5 years	50,000	833
• Feed cost for 5 years	3,10,250	5,171
• Labour cost for 5 years	36,000	600
• Housing cost for 5 years	12,500	208
Total expenditure for 5 years	3,58,500	5,975
Total income from selling milk in 5 years	5,11,000	8,517
Total income from selling manure	25,000	416
Income from buffalo after 5 years	1,27,250	2,120
Total selling price of unproductive buffalo (for meat purpose) after rearing for 5 years (depending on live weight (approx 200 kg))	10,000-11,000	1,66-183

Source: FICCI primary survey

As per the findings of the survey, farmers are able to sell animals at better prices in livestock markets. Most animals are transported to the livestock market by tempos or vans, but a few animals are brought to market on the hoof from nearby villages. Entrance fees to the livestock market if any are also borne by the farmers. For instance, in Maharashtra an entrance fee of approximately ₹10/animal is charged for entering a livestock market.

(2) Cost incurred by traders:

Table 3. Expenses incurred by traders in domestic value chain

Particulars		₹ per head	USD
Cost of buffalo (based on estimated live weight [150-200kg])		8,000-10,000	133-166
Average variable expenses of trader after purchase of buffalo		₹ per head	USD per head
1	Transportation cost to municipal slaughter house	100-150	1.66-2.50
2	Market fees at livestock market (animal mandi)+ taxes	50-100	0.83-1.66
3	Fees for slaughtering at municipal slaughterhouse	30	0.50
4	Fees for health certificates obtained from veterinarians	20	0.33
5	Total cost per animal (approx)	200	3.33
		₹ per kg.	USD per kg.
5	Wholesale (Bone-in) price of buffalo meat to retailer (Feb- May 2013)	115	1.92
6	Wholesale (Bone-in) price of buffalo meat to consumer (Feb- May 2013)	130	2.1

Source: FICCI primary survey

- 1. Cost of buffalo:** The price of a buffalo is decided on the basis of the estimated weight and age of the animal. Variation is noticed in purchase prices of buffalo bought from livestock markets and those bought from a farmer's farmstead. For instance, a trader can buy a buffalo weighing 200 kg at a price of ₹10,000 from a farmstead and can buy the same animal from a livestock market for ₹11,000.

Table 4. State wise average carcass yield and breeds of Buffalo

		Avg carcass yield (kg) (bone in carcass)	Breeds
		kg/animal	
1	Maharashtra	139	Mehsana, Murrah, Nilli, Surti, Nondescript
2	Punjab	135	Murrah, Nilli, Ravi
3	Haryana	–	Murrah, Ravi
4	Uttar Pradesh	130	Murrah, Nilli, Ravi
5	Andhra Pradesh	103	Nondescript, Murrah, Surti, Mehsana
6	Kerala	102	Nondescript
7	Karnataka	108	Murrah, Nondescript
8	Uttarakhand	125	Murrah, Nilli
9	Rajasthan	70	Nondescript, Murrah, Nili, Ravi, Surti
10	Bihar	62	Nondescript

Source: Basic Animal Husbandry Statistics (DADF), Ministry of Agriculture, Government of India & Central Institute for Research on Buffaloes

2. **Transport costs:** Given the regional distribution of farmers and distances to market, transport cost is a major component of the value chain. Transport cost is borne by the livestock trader, if he purchases the animal directly from farmsteads. If the trader buys from a livestock market, he has to bear the transport cost only to the municipal slaughterhouse. The transportation charges from mandi to slaughterhouse depend on the distance and number of animals that can be accommodated in a truck. A minitruck has the capacity to transport a maximum of two animals. However, livestock traders use big trucks if the number of animals is between 10 to 15. The average transport cost incurred by traders per head is ₹ 100-150. A veterinarian issues a veterinary health certificate (fit for transport to slaughter house) for the transit of the animal from the livestock market to the slaughterhouse.
3. **Market Fees:** The custom of weekly markets is very popular in India and has been sustained for decades. Livestock markets play an important role in assembly and sale of livestock. In the animal trade between the farmers who rear the animals and the butchers a minimum of one to two middlemen are involved who use these markets as their platform for transactions. These livestock markets can be classified into two types: secondary assembly markets and distribution markets.

Weekly livestock markets (secondary assembly markets) are so called because they are held on a specific day of the week. Such weekly markets are highly prominent in livestock trade, especially for small and marginal farmers. Animals are brought to these weekly markets by petty traders. Such markets are under the control of local bodies.

Distribution markets are located adjacent to urban slaughterhouses. Traders transport animals either from weekly markets or directly from villages to these markets and sell the animals to butchers or wholesale meat dealers, through commission agents.

Table 5. Major livestock markets

S. No	Place	Livestock	Markets
1	Gudur, Nellore, AP	Sheep And Goats	Weekly market(Fridays)
2	Pollachi, Coimbatore District, Tamil Nadu	Cattle and Buffalo	Biweekly(Tuesday, Thursday)
3	Deonar Abattoir campus, Mumbai	Large animals, small animals	Biweekly (Monday and Friday) (Tuesday and Thursday)
4	Ghoti, Nasik	Large animals, small animals	Biweekly(Monday and Tuesday)
5	Pundooah, Hooghly dist, West Bengal	Large animals, small animals	Biweekly (Sunday and Wednesday)
6	Lokhra and Maligaon markets, Guwahati, Assam	Swine	Lokhra on Sunday and Maligaon on Monday

Source: Meat & Poultry Processing Board, Ministry of Food Processing Industries, GOI, report on "Socio Economic Upliftment of Meat Industry Workers" prepared by NABARD Consultancy Services (NABCONS), 2013

Some agricultural marketing departments of the states have brought marketing of livestock under the Agricultural Produce Markets Acts (APMA) notified by various states. All market actors, particularly those maintained by local authorities or the state Governments governed by APMA, are licensed and an annual fee is collected from them. In most markets, transactions take place after examination of the animals by the buyers, through brokers/commission agents. A market fee is paid by purchaser to suppliers post transaction in the market area. Huge variation is noticed in the amount of market fees paid in different states of India.

Table 6. Market fees at livestock markets in different states

	State	Market fees
1	Punjab	1% of the cost of animal
2	Andhra Pradesh	₹ 100-200/head
3	Maharashtra	₹ 10/head/day: municipality charges, ₹ 50 stamping fees/animal, ₹ 15 mandi fees
4	Delhi	₹ 150 per truck (1 truck can haul 5-20 buffalo), average 10 buffalo
5	Rajasthan (Jaipur)	₹ 35/head
6	Uttar Pradesh (Bareilly)	₹ 50/head
7	Uttar Pradesh	₹ 20-50/head

Source: FICCI primary survey

- 4. Slaughtering fees:** A slaughter fee is charged by municipal corporations/panchayats/town committees for the facilities provided for slaughtering of buffalo.

Table 7. Slaughtering fees at municipal slaughterhouses in different states

		₹/head	USD/head
1	Maharashtra (Mumbai)	210	3.50
2	Jaipur (Rajasthan)	35	0.58
3	Uttar Pradesh (Saharanpur)	18	0.30
4	Uttar Pradesh (Bareilly)	45	0.75
5	Delhi	300	5.00
6	Andhra Pradesh (AP)	20	0.33
7	Kerala	200	3.33
8	Karnataka	50	0.83
9	Punjab	No municipal slaughterhouse for buffalo	

Source: FICCI primary survey

The fees vary from a minimum of ₹20/head to a maximum of ₹300/head. This fee is paid by the trader or butcher who brings buffalo for slaughter. A few modern municipal slaughterhouses, such as Gazipur municipal slaughterhouse, also provide facilities for chilling of meat. Debonair abbatoir in Maharashtra provides a facility of transporting meat from the abbatoir to retail shops, for which it charges a fee of ₹100 per head.

Table 8. Margin in buffalo meat trade between livestock trader and retailer, in ₹

Approx weight of buffalo considered= 200 kg						
Margin	Head	Hide	Feet	Stomach	Red offal	Meat
Livestock trader to retailer	100	1000	15	15	120	₹ 115/kg (avg carcass 110 kg)
Retailer to consumer or end user	120	1100	20	20	150	₹ 130/kg (avg carcass 104 kg due to 5% dockage)

Source: FICCI primary survey

In 2005 the prices of buffalo meat in domestic market generally ranged between ₹45 and 50 per kg. Prices increased to ₹90 to 95 per kg in 2012 and there has been a further increase in prices in 2013, to ₹130 per kg. With further growth in demand, meat prices are likely to increase in future. Retail prices of buffalo meat as collected during the survey are shown below.

Table 9. Retail price of buffalo meat in domestic market

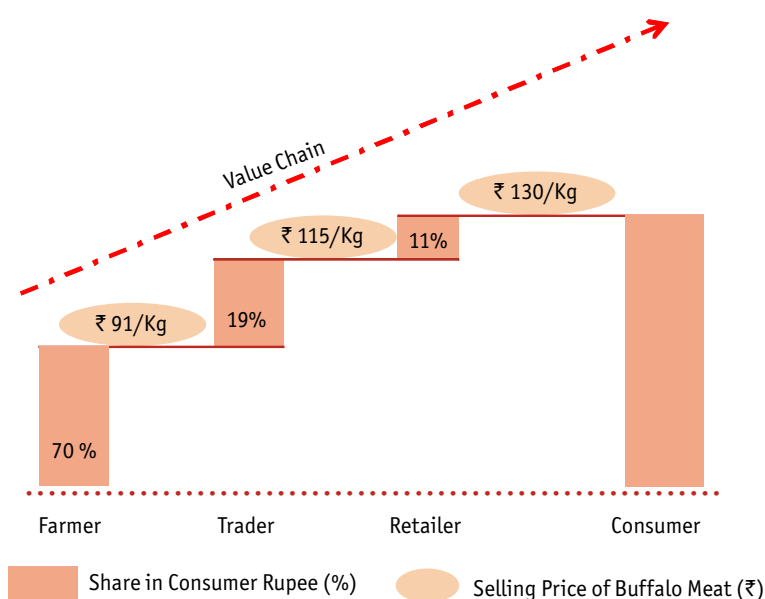
	State	Retail price of buffalo meat (₹/kg), (May- June 2013)
1	Andhra Pradesh	150-160 (bone in)
2	Bihar	120 (bone in)
3	Delhi	160 (bone in), 200 (boneless)
4	Karnataka	160 (bone in), 175 (boneless)
5	Kerala	180 (bone in), 200 (boneless)
6	Rajasthan	105 (bone in)
7	Uttar Pradesh	120-130 (bone in)
8	Uttarakhand	120-130 (bone in)
9	West Bengal	120 (bone in)

Source: FICCI primary survey

Buffalo Meat Value Chain for Domestic Market: Share in Consumer Rupee

The figure below describes various stakeholders involved in the marketing channel of buffalo meat production for the domestic market and also the aggregate value realisation by them in consumer rupees. The farmer gets 70% of the aggregate value realisation in consumer rupees, whereas the trader (middlemen) who generally sells the meat at ₹115 per kg earns 19% of the share. The retailer claims 11% of the balance.

Figure 4. Buffalo meat value chain for domestic market: Share in consumer rupee



Note: Weight of Buffalo: 200 kg, Carcass Yield:110 kg, Approximate Selling Price of Buffalo: ₹ 10,000/head

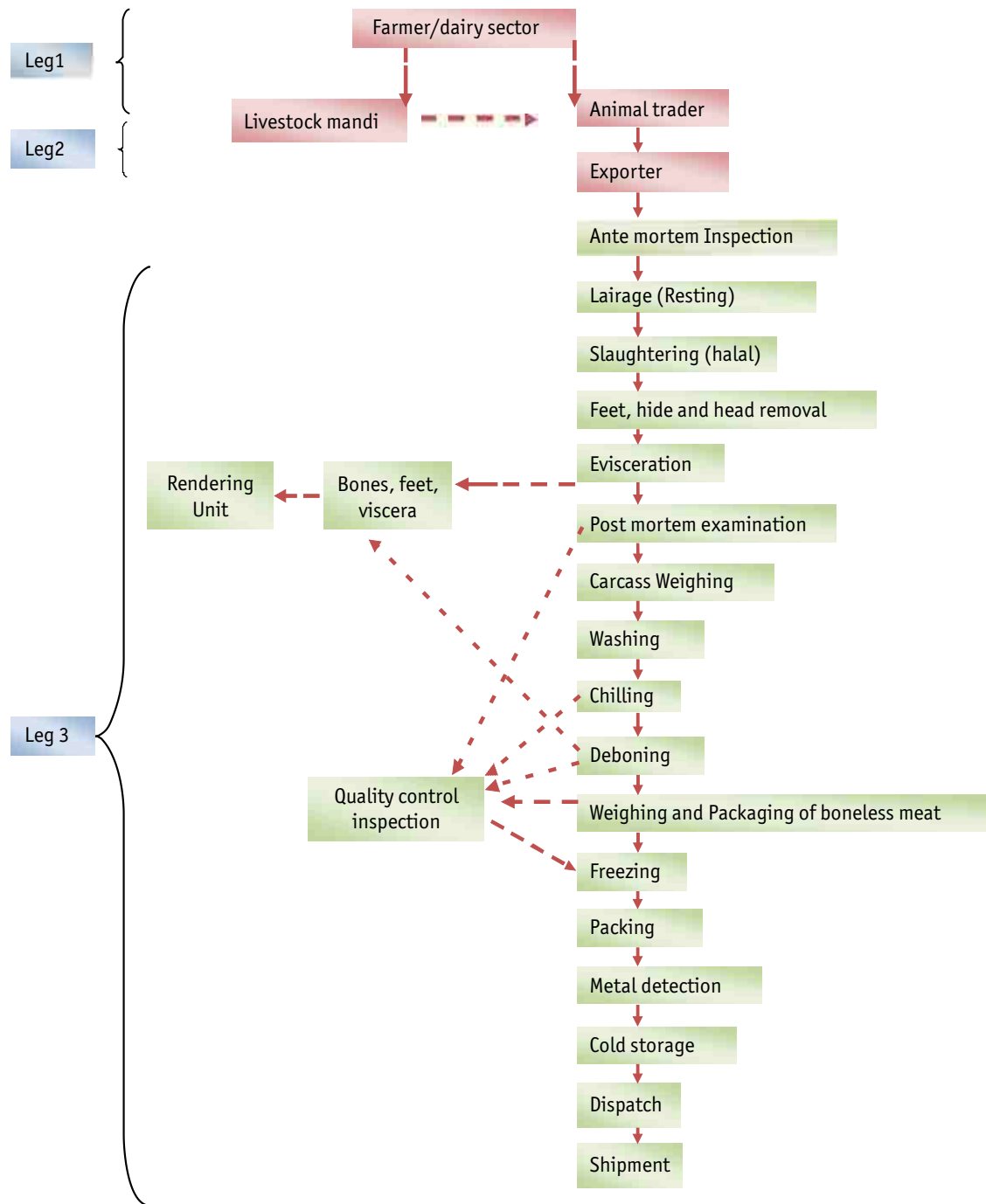
Source: FICCI primary survey

In view of the importance of buffalo in India and their contribution to the rural economy in many ways, these ruminants deserve due attention in terms of policies, programmes and health care. There is need for improvement in production, postproduction and consumption systems, including the viability of establishing municipal slaughter houses. The recommendations for developing a competitive Indian buffalo meat value chain for domestic market are given in chapter 7.

Section 5.B

VALUE CHAIN OF BUFFALO MEAT PRODUCTION FOR EXPORT MARKET

Figure 5. Value chain of buffalo meat production for export market



The buffalo meat value chain for export has different stakeholders working at several stages of production. Farmers raise the buffalo till they become uneconomical and thereafter sell them to traders. Traders play the role of aggregators in the value chain and sell the buffalo to exporters for further processing and sale in export markets.

Section 5.B.1

Stages in export of value chain and roles of different stakeholders

Leg1: Farmers rear buffalo for milking and breeding and cull them for meat production when the animals becomes unproductive. Only unproductive buffalo are utilized for meat exports.

The buffalo are reared primarily by farmers in small numbers. Farmers rear buffalo in basically three stages: buffalo-calf, growing and feeding. The livestock used for meat exports are derived from unproductive, culled animals at the end of their productive lives. Only animals permitted under the Animals Preservation Acts are slaughtered for exports. According to the Prevention of Cruelty to Animal Slaughterhouse Rules (2001) no animals may be slaughtered which are

- (i) pregnant, or have offspring less than three months old, or
- (ii) are under the age of three months, or
- (ii) have not been certified by a veterinarian that they are fit for slaughter.

Farmers sell buffalo to traders or at a livestock market. The price received by farmers is based on the estimated weight of the animal as shown in Table 10.

Table 10. Farmgate price of live buffalo for export market, ₹/head

	Estimated liveweight per head	Carcass yield, kilograms	Livestock market prices, ₹/head	Trader prices, ₹/head
1	400	240	24,000	23,000
2	350	210	20,000	19,000
3	300	180	17,000	16,000

Source: FICCI primary survey

Leg2: Livestock traders play crucial role as assemblers and suppliers of buffalo in quantity to exporters.

As demand for buffalo meat is increasing, many exporters are feeling the need to forge better linkages for an interrupted supply of buffalo for meat production. Exporters are not linked to farmers directly, however. This has given rise to a class known as 'traders' who collect buffalo from different farmers directly or buy animals from livestock markets and sell to exporters. Livestock traders are paid weekly. The payment is decided on the basis of bone-in carcass weight after post mortem examination.

Table 11. Wholesale prices of buffalo paid by exporters

	Liveweight, kilograms	Carcass yield per head, kilograms	Prices paid by exporters, ₹/head Price per kg= ₹122-125/kg bone-in
1	400	240	29,280
2	350	210	25,620
3	300	180	21,960

Source: FICCI primary survey

Leg3: Processing at exporter's unit and selling in international market

Processing stage at exporter's unit: After receiving, buffalo are kept in lairage for 24 hours for inspection by State Government veterinary officers. The first processing stage starts with slaughter of the buffalo at an abattoir. The supplier (trader) is paid on the basis of carcass weight. The carcass is converted to boneless meat and bones. Bones, inedible viscera and feet are sent to the rendering unit, where meat and bone meal is produced for use in poultry feed. Hides are sold to the leather tanning industry. Edible viscera, heads, and feet are also sometimes sold in the domestic market. Boneless meat is packaged in food-grade polythene film, frozen (at -18°C), packed in cartons and kept in cold storage for export. Cartons are dispatched in refrigerated containers for shipment.

Table 12. Edible and inedible offals

	Edible offals	Inedible offals
1	Heart	Oesophagus
2	Liver	Stomach
3	Kidney	Intestines
4	Spleen	Hide
5	Lungs and Trachea	Hooves
6	Tongue	
7	Brain	
8	Muscle of Head and Feet	

There is no strict classification of edible and inedible offal. Inedible offal can be converted to edible or edible offal can be treated as inedible offal in some instances.

Section 5.B.2 EXPENSES INCURRED BY EXPORTERS

Major costs involved in undertaking primary operations in export units after receiving of animals and shipment of final marketable products include (a) transportation, (b) packaging, (c) service costs, (d) electricity and diesel, and (e) labour. In the entire value chain, transport costs, which consume nearly two fifths of the total expenditure, end up being the most expensive operation category.

Figure 6. Expenses incurred in different operations at exporters end

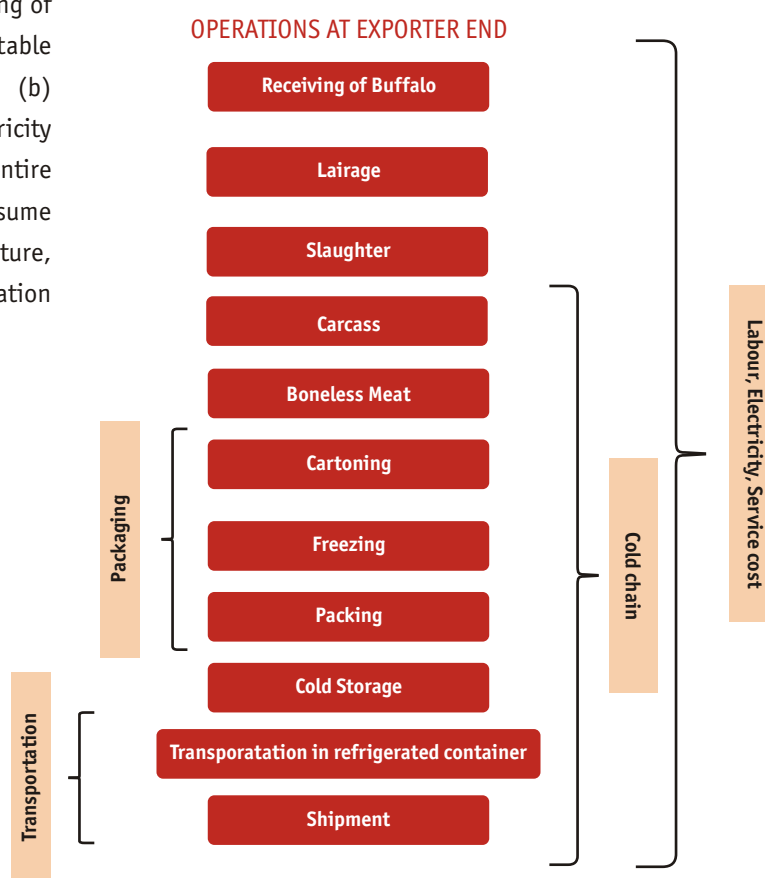
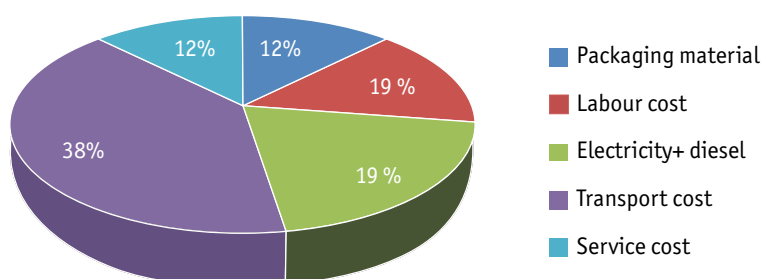


Figure 7. Expenses incurred by exporter in production of buffalo meat*



* does not include cost of buffalo
Source: FICCI primary survey

(A) Packaging cost:

The estimated packaging cost for 20 kilograms of boneless meat is given in Table 13. The packaging material includes (a) food grade polythene-lined cardboard carton, (b) polythene used for shrink wrapping, and (c) straps used for wrapping. While estimating the cost of packing material, ordinary packaging has been taken into consideration. However, to enhance the shelf life of meat and its quality, vacuum packing/MAP (modified atmospheric packaging) may also be used. The cost of such a packing will be additional.

Table 13. Cost of packaging of 20 kilograms of boneless meat

	Particular	Amount	Cost, ₹
1	Polythene	75 grams	4.50 per 75 gm polythene
2	Carton	1	19 per carton(one carton can carry 20 kg meat)
3	Strap		0.50
4	Total cost		1.5 - 2.0 per kg meat

Source: FICCI primary survey

(B) Labour cost:

Labour is another important cost incurred by exporter. It has 2 components, skilled and unskilled labour. Skilled labour includes butchers, packers, checkers and supervisors. Unskilled labour includes labourers for cleaning, washing, tending to buffalo in lairage, loading and unloading of meat cartons. Butchers slaughter and skin the buffalo, split the carcasses, and debone. Careful selection and carving of prime cuts (boneless eg: tender loin, strip loin, blade, forequarters etc) of the buffalo carcass is a major skill required in butchers.

Packaging of meat is usually done by females. Checkers are responsible for selection of meat cuts, whereas supervisors play a role in monitoring the staff during processing as well as the slaughter floor. A slaughterhouse with production capacity of 700 head per day runs two shifts.

Table 14. Labour requirement for an abattoir of production capacity of 700 animals/day

	Type of labour	Salary per month (₹)	Number	Function
(A)	Skilled			
A.1	Deboning butchers	12,000	90	Separation of bones and flesh
A.2	Slaughter butchers	12,000	30	Slaughtering operations
A.3	Meat packaging staff: female labour	6,000	22	Polythene packaging
A.4	Carton packaging	6,000	20	Carton packaging and strapping
A.5	Checkers	12,000	18	Sorting of meat cuts

	Type of labour	Salary (₹/month)	Number	Function
(B)	Unskilled			
B.1	Animal attendants	5000	4	Handling of animals at lairage
B.2	Sanitation staff	5000	20	Cleaning and washing of plant
B.3	By product handlers	5000	5	By product handling

Source: FICCI primary survey

(C) Transport cost overland, average distance 1500 km (plant to port of shipment, North India plants to Western Indian ports of shipment):

Transport cost includes the cost of hauling boxed meat from the packing plant to the port of shipment. Mode of transport used is refrigerated container, which is hauled overland by road or rail. During transport the cold chain (temperature below -18 degree Celsius) is maintained. Container capacity is 28 tons of meat.

(D) Electricity cost:

Major technical operations in abattoirs require power, for running the lines, lighting, and chilling, as well as operating compressors for freezing and cold storages, sterilizers, packaging machines, etc. Electricity cost is high due to high tariff, generation of power by generators during power cuts, voltage fluctuation and load shedding. On the other hand the large power requirements for maintaining cold chain (air conditioning, carcass chilling, meat freezing at -18°C) adds to overall electricity costs.

(E) Service cost:

The exporter bears the cost of services, which include maintaining quality assurance, shipping agents' fees, certifications, and laboratory analyses. These costs vary from state to state. In some states the exporter is charged for services of antemortem and postmortem inspection in addition to certification by the veterinary department of the state Government, whereas in other states only veterinary certification costs are charged. However, on average this cost comes to 1 to 1.5 ₹ per kg of meat exported.

Table 15. Service cost for export unit of production capacity of 700 buffalo/day

		Annual charges, ₹	USD
1	Food safety and quality Certifications		
	HACCP certification	1,00,000	1,667
	ISO certification	1,00,000	1,667
	FASSI certification	50,000	833
	APEDA certification	50,000	833
2	Microbiological and Chemical testing centre	10,00,000	16,667
3	Water and meat testing (lab accredited by NABL)	1,00,000	1,667
4	Training of skilled workers and technical staff (in HACCP)	50,000	833
		Per container, ₹	
5	Commission to shipping agent	20,000	333

Source: FICCI primary survey

(F) Capital Cost

Estimated capital cost for constructing an export unit of capacity 700 head/day is shown in the table below

Table 16. Capital cost for constructing an export unit with production capacity of 700 head/day

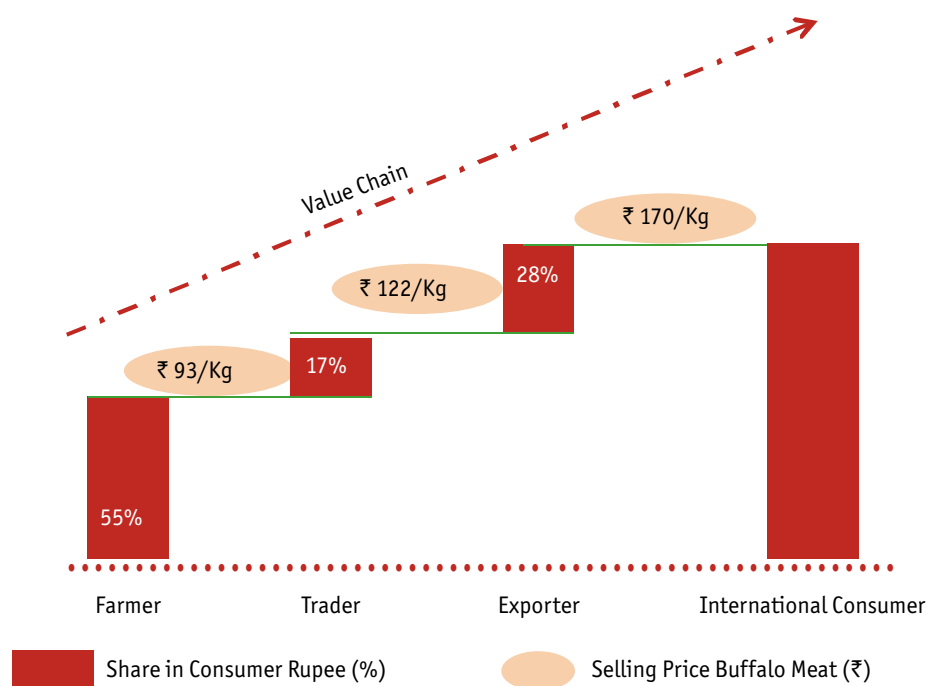
Particular	₹	USD
1. Civil work	12,00,00,000	2,000,000
2. Cold store equipment	4,00,00,000	666,666
3. Rendering equipment	8,00,00,000	1,333,333
4. Slaughterhouse equipment	10,00,00,000	1,666,666
5. Utilities (Land and ETP not included)	5,00,00,000	833,333
TOTAL	39,00,00,000	6,500,000

Source: FICCI primary survey

Buffalo Meat value chain for Export Market: Share in Consumer Rupee

The figure below describes various stakeholders involved in the marketing channel of buffalo meat production for export market and also the aggregate value realisation by each of the consumer rupee. The farmer gets 55% of the aggregate value realisation, whereas the trader (middlemen) who generally sells the meat at ₹ 122 per kg to the exporter, earns 17% of the share. The exporter who adds value in the supply chain by undertaking buffalo meat processing in a fully integrated modern abattoir with rendering unit and effluent treatment plants (including the expenses made by exporters on overall operation, electricity, packaging, labour, transport, service etc. as mentioned in Figure 7) gets 28% of the share.

Figure 8. Buffalo meat value chain for export market: Share in consumer rupee



Note: Weight of Buffalo: 300 kg, Carcass Yield:180 kg, Approximate Selling Price of Buffalo: ₹ 17,000

Source: FICCI primary survey

Buffalo in India contributes about 30% of total meat production in the country. The livestock rearing constitutes an important vocation in India, next only to crop raising. In addition to providing milk and manure, they serve as a major source of animal protein for the millions of people (in domestic and international markets) and animal by-products, which if handled properly can earn sizeable foreign exchange in addition to meat. The recommendations for developing a competitive Indian buffalo meat value chain for export market are given in chapter 7.



6

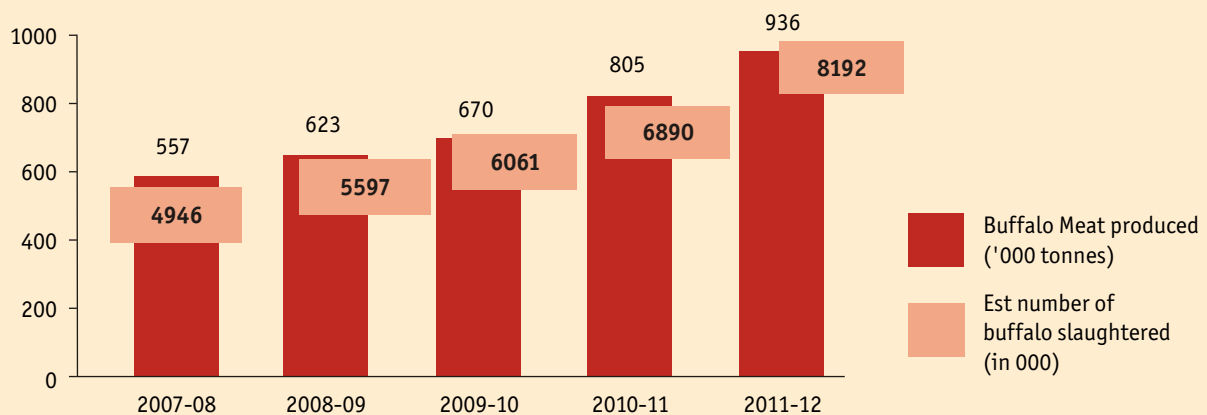
India's Production Capacity and Demand for Buffalo Meat in the Domestic and International Markets

India's Production Capacity and Demand for Buffalo Meat in the Domestic and International Markets

6.1. India's production capacity

The buffalo in India, apart from providing milk, contribute immensely to the national economy. According to data compiled by the animal husbandry departments of all states, buffalo meat produced in registered municipal slaughterhouses increased from 5,57,000 tonnes in 2008 to 9,36,000 tonnes in 2011. Uttar Pradesh was the top buffalo meat-producing state in 2011. Details of the production capacity of major states in India are given in the annexure. As per the findings of the survey and insight into the buffalo meat production data (source Animal Husbandry Statistics, GOI, Ministry of Agriculture, DADF) production figures of buffalo meat do not possibly include production for export.

Figure 9. Buffalo slaughter numbers and carabeef produced in India for domestic consumption



Source: Basic Animal Husbandry Statistics, DADF, Ministry of Agriculture, GOI, State / UT Animal Husbandry Departments

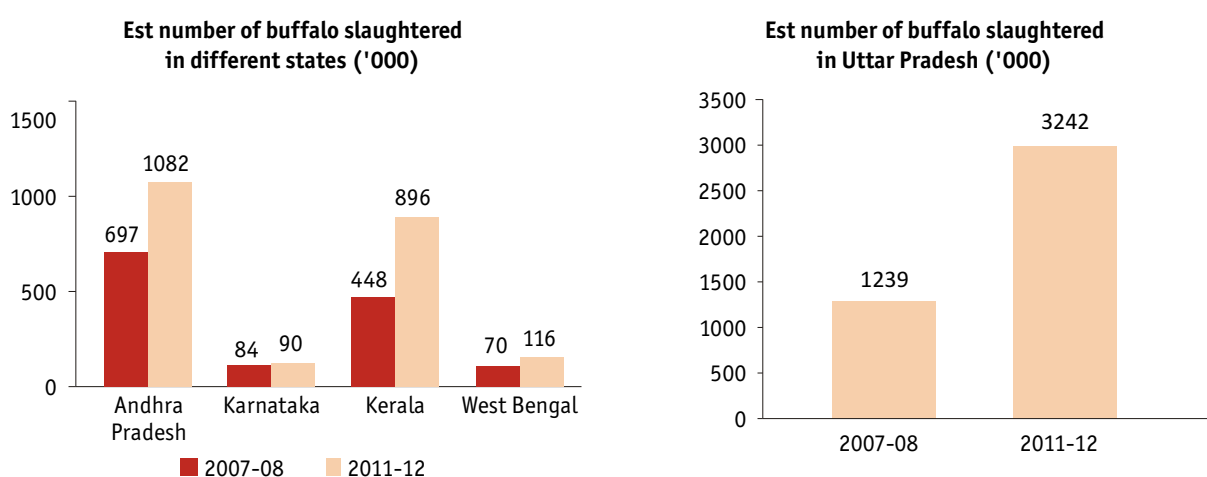
*As per the response received from various state government officials, this may not necessarily include export figures of all the states. This includes meat produced in municipal slaughterhouses or registered private slaughterhouses for domestic consumption.

6.2. Domestic market of buffalo meat in India.

Most meat consumed in India is produced in municipal slaughterhouses. Facilities for slaughtering small ruminants (sheep and goats) are generally found in small towns or rural areas, whereas municipal slaughterhouses for large ruminants are available in large cities only. There are 3600 municipal slaughterhouses in India which slaughter animals for domestic consumption.

The major consuming states of buffalo meat in India are Uttar Pradesh, Andhra Pradesh, Kerala, West Bengal and Maharashtra. The team visited major municipal slaughterhouses in these shortlisted states to understand the pattern of buffalo meat demand in the domestic market. The rising slaughter of buffalo in these states is indicative of increasing consumer demand for buffalo meat. We can assume that buffalo meat produced in a majority of municipal slaughterhouses in India is used for domestic consumption. This emerges from the fact that there are very few slaughterhouses in India authorised by APEDA to supply buffalo meat for export. As a matter of policy, meat exported from India should be sourced from abattoirs and meat processing plants registered with APEDA. For example, the Gazipur municipal slaughterhouse at Delhi, presently leased to Allana, is authorised for this purpose. However, such municipal slaughterhouses are very few although numbers are steadily increasing post the Policy wherein approval of APEDA for an export abattoir has become mandatory.

Figure 10. Trends in increasing consumption of buffalo meat in the domestic market



*Increasing trend of buffalo slaughter in municipal slaughter houses in mentioned states indicates increasing domestic consumption.

Source: Basic Animal Husbandry Statistics (DADF), Ministry of Agriculture, Government of India

Trends in buffalo meat consumption in domestic market.

- I. Slaughter of buffalo for domestic consumption is trending upward in some major meat producing states, which points to increasing domestic demand.
- II. The pattern of buffalo meat consumption in India may differ from state to state due to differences in consumer preference and religious taboos. Buffalo meat in India is generally consumed by the Muslim community and in some places by Christians as well. Therefore, growth in the population of buffalo meat consumers will lead to a direct increase in domestic demand for buffalo meat.

- III. Buffalo meat is one of the cheaper options of animal protein for consumers. The average price of buffalo meat is ₹ 130-150/kg whereas the average price for mutton is ₹ 380/kg, fish is ₹ 200-500/kg, and chicken is ₹ 180 per kg.
- IV. There are no restrictions on buffalo slaughter when compared to cows (genus *Bos* spp.) in most states. Laws affecting slaughter are described in an annexure.
- V. Being relatively low in fat and cholesterol, buffalo meat is one of the most healthful red meats for Indians.

Functioning of slaughterhouses in India:

Slaughter takes place at stated hours in municipal slaughterhouses and the meat is examined by Government appointed veterinary officers. The slaughter of food animals in India is a state subject, wherein local bodies such as municipal corporations, municipalities and town panchayats are the main agencies. These local bodies collect slaughtering fees from meat traders and retail meat stall owners for providing the facilities for slaughter and dressing of animals. Besides providing facilities for slaughter, municipal slaughterhouses also provide for antemortem and postmortem inspection by state veterinary officers. Sanitation staff are also provided for day-to-day cleaning and sanitation work. State Governments and union territories receive financial aid from the union Government to improve existing slaughterhouses and set up new abattoirs.

Table 17. Production capacity (head/day) of major municipal slaughterhouses of states surveyed

	State	Municipal Slaughterhouse (MSH) visited	Approx Buffalo slaughtered/day, head
1	Uttar Pradesh	Rampur MSH	150
		Bareilly MSH	400
		Aishbagh MSH	300
2	Andhra Pradesh	Chengicharala MSH	300
3	Maharashtra	Debonair Slaughterhouse	200
4	Delhi	Gazipur Slaughterhouse	1000
5	West Bengal	Tangra MSH,	450
		Garden Reach MSH	150-200
6	Rajasthan	Chainpura MSH	800-1000
7	Karnataka	Civil Slaughterhouse, Bengaluru	150

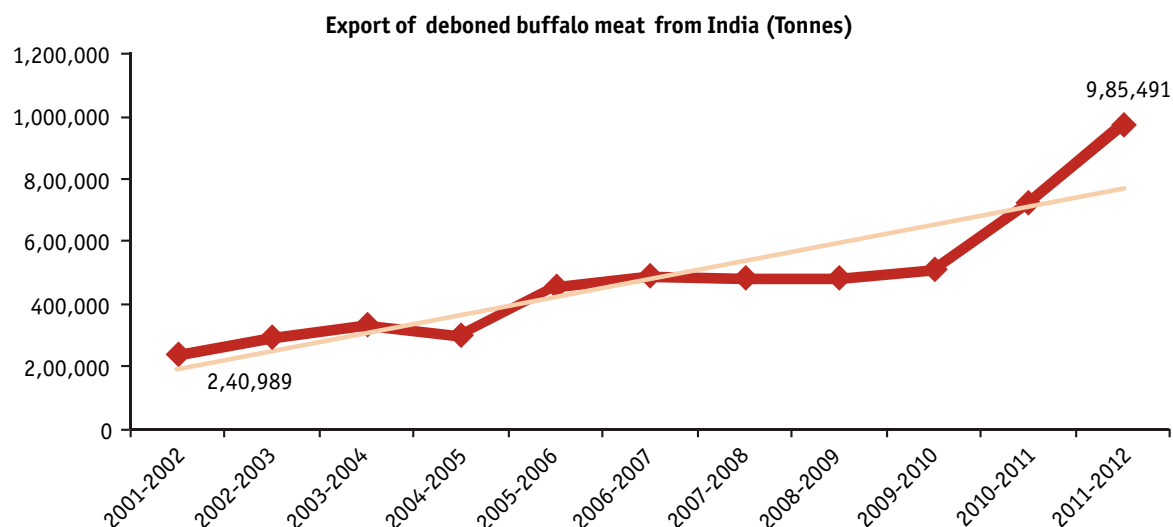
Source: FICCI primary survey

Note: These states are also the major consumers of buffalo meat in India

MSH=Municipal Slaughterhouse

6.3 Export Market of Indian Buffalo Meat

Figure 11. Export of deboned buffalo meat from India



Source: APEDA

Export of buffalo meat has emerged as an important component of India's agricultural exports, underpinned by strong global import demand for buffalo meat. Buffalo meat valued at ₹17,400.59 crores (USD 2.9 billion) was exported in FY 2012-13. India has a competitive advantage in the export of buffalo meat. Besides originating in a source strategically located near meat importing countries, Indian buffalo meat offers many positive characteristics.

Strengths of Indian buffalo meat for export market

1. The livestock in India are reared on green pastures and agricultural crop residues, thus are raised under a green livestock production system.
2. Conducive Government policies
 - Licensing and registration of meat packing plants/abattoirs: Government of India has laid down standards for export of meat which includes standards for abattoirs, processing plants for various meat products. Registration of abattoirs and meat processing plants is done by the Agricultural and Processed Food Products Export Development Authority (APEDA).
 - Government of India accords high priority to quality issues in buffalo meat for export. Therefore a stringent regulatory mechanism has been put in place by the Government in the form of mandatory approvals of modern abattoirs for production of buffalo meat for export. APEDA is also enforcing HACCP and approves the plants with HACCP accreditation through an interdepartmental panel. Meat of genus *Bos* spp. is banned for export. In recent years the Government has taken note of this and promulgated regulations requiring the exporter to declare the source of meat. However, there is

need to strengthen certification procedures to ensure only buffalo meat as per capacity approved by APEDA is exported from each approved unit.

- **Animal Health Certificate:** According to the current Export and Import Policy of the Government of India, each export consignment is subject to compulsory microbiological and other tests and a comprehensive pre-shipment inspection certificate is issued by a Government laboratory. Each consignment is accompanied by this health certificate. This certificate also states that the meat has been prepared from healthy livestock, is free from contagious and infectious diseases, including foot-and-mouth disease. The health certificate also confirms that the livestock have been subjected to antemortem inspection followed by postmortem examination, and that the meat is fit for human consumption. In fact the procurement of animals from registered markets to the final exports of the meat consignment is under the constant monitoring and inspection of Government agencies, besides having the services of highly qualified and experienced veterinarians, microbiologists, etc.

Meat consignments conforming to the notified standards and specifications are allowed for export. Each consignment is accompanied by a test certificate issued by any of the above agencies, based on the microbiological examination of the consignment

3. India is a member country of the International Organization for Animal Health (OIE), and is mandated to report list "A" and list "B" animal diseases to the OIE at regular intervals. Among the list "A" disease of cattle and buffaloes, except for foot and mouth disease, India is free from all other bovine diseases, namely rinderpest, contagious bovine pleuropneumonia, lumpy skin disease, etc. In the case of FMD, central and state government are taking serious control measures. For control of FMD, 54 districts had been identified where 100% vaccination was completed during the 10th Five Year Plan (2002-2007). In the 11th Five Year Plan (2007-2012), the FMD control programme is on going in these 54 districts and additional districts have also been included to a total of 230. The animals are vaccinated every six months under the programme.
4. There is no practice of using hormones, antibiotics or any other chemicals to promote growth and fattening of livestock.
5. Indian buffalo meat is 93% chemically lean and blends very well with other ingredients for value added products.
6. The animals are strictly halal slaughtered.
7. Indian buffalo meat is low in fat and cholesterol.
8. Indian buffalo meat is free from radiation.

India caters to the meat demand of major Islamic countries as the meat is genuinely halal, lean and nutritious, more healthy than beef. Indian buffalo meat is ideal for processing since it retains more moisture and binds easily with ingredients. Indian buffalo meat is witnessing strong demand in international markets due to its lean character and organic nature. Also India has a cost advantage as Indian buffalo meat is most competitively priced red meat in the world.

Table 18. Buffalo Meat Export Price (2009 - 13)

Year	₹/Kg	USD\$/Kg
2009-10	112.18	1.88
2010-11	119.53	1.99
2011-12	142.5	2.37
2012-13	170-180	2.833-3

Source: DGFT 2012 (till 2011-12), FICCI survey 2013

Meat Export Policy

The Meat Export Policy was further strengthened in the year 2004 when it was notified (vide notification no 12/ (2004-2009) dated 21 December, 2004) that export of meat and meat products will be allowed subject to the exporter furnishing a certificate to Customs at the time of export that these items have been obtained/sourced from an abattoir/meat processing plant registered with APEDA. The policy was further strengthen in 2011, by issuing a more comprehensive notification (Notification No 829RE-2010)/2009-2014 dated 31 October, 2011) that

Exporters are required to certify that

- The items have been obtained/sourced from an APEDA registered integrated abattoir or from APEDA registered meat processing plant, and
- That the raw material has been sourced exclusively from an APEDA registered integrated abattoir.

There are established rules on cruelty to animals and transportation which are required to be followed by the concerned under the Prevention of Animal Cruelty Act, 1960, and the Transport of Animal Rules 1978. Under these rules, Section 56, sub section (c), no goods vehicle shall carry more than six cattle, besides other restrictions were also imposed. These rules were amended on September 22, 2009 by the Ministry of Environment and Forests to include a provision that animal movement is subject to a minimum space requirement per head in the truck. Larger vehicles, e.g., those with floor area of 16.54sq. m., can carry 16 cattle of average weight up to 200 kgs, 14 cattle of average weight of up to 200-300kgs, or 12 cattle of average weight of 300-400kg.

The Ministry of Commerce periodically reviews the meat export policy and takes necessary steps considering all stakeholders' and national interests:

- In 2006, the policy was amended to ensure that all packages of buffalo meat for export display requisite details to ensure traceability/minimize malpractice.
- In 2006, the policy was also amended to prohibit export of bone-in buffalo meat to increase product safety.

- In 2011, the policy was amended to ensure sourcing of raw material for buffalo meat export only from abattoirs and meat processing plants, having an APEDA Plant Registration Certificate.
- In 2011, the policy was amended to enable veterinarians registered under the Indian Veterinary Council Act, 1984 to inspect/conduct ante mortem inspection and product testing at APEDA certified labs to enable the competent authority to issue health certificates.

Table 19. Foreign Trade Policy for Export of Bovine Meat

S. No	Tariff HS Code	Item Unit	Item description	Export Policy	Nature of restriction
18	0201 1000	Kg	Beef of cows, oxen and calves	Prohibited	Not permitted to be exported
	0201 2000				
	0201 3000				
	0202 1000				
	0202 2000				
	0202 3000				
	0202 3000				
19 (a)	0201		Meat of buffalo (both male and female), fresh and chilled	Prohibited	Not permitted to be exported
	0201 1000				
	0201 2000				
	0202		Other bone-in cuts		
	0202 1000				
	0202 2000				
			Meat of buffalo (both male and female), frozen)		
	Other cuts with bone				
19 (b)	02013000	Kg	Boneless meat of buffalo (both male and female), fresh and chilled	Free	1. Export allowed on production of a certificate from designated veterinary authority of the state from which the meat or offals emanate, to the effect that meat or offals are from buffalo not used for breeding and milch purposes 2. Quality control and inspection under note 3 and 4 respectively as well as condition stipulated under note 6 are fulfilled.
	0202 3000		Boneless meat of buffalo(both male and female), frozen		

Source: APEDA, Indian meat industry, Red Meat manual



7

Conclusion

Conclusion: Issues and Recommendations in a Nutshell

Recommendations for domestic value chain

1. There is need for a sustainable buffalo production system in the country to meet the future demand of buffalo meat. Therefore priority should be given to implementation of a **male buffalo calf rearing programme for the long-term benefit of the farmers, country and the trade.**

Although India ranks at the top in bovine inventory, the inventory of buffalo has been found to be declining in a few states. This indicates a growing shortage of buffalo meat. However, such a decline cannot be related to the export of buffalo meat from the country as there are no registered export units in these states. Secondly, slowdown in increase of population is not only due to slaughter but can also be due to urbanization and conversion of agricultural areas into housing/commercial/industrial land.

Table 20. States with decline in buffalo population

State	Pop. In 2003 (in thousands)	Pop. In 2007 (in thousands)	Annual growth rate (%)	Number of export units
Assam	678	500	-7.33	Nil
Haryana	6035	5953	-0.34	Nil
Himachal Pradesh	774	762	-0.40	Nil
Kerala	65	58	-2.75	1
Odisha	1394	1190	-3.88	Nil
Uttarakhand	1228	1220	-0.17	Nil
West Bengal	1086	764	-8.42	Nil

Source: Basic Animal Husbandry Statistics, DADF, Ministry of Agriculture, GOI, State / UT Animal Husbandary Departments

Small and marginal farmers play a major role in buffalo husbandry. An analysis of survey results shows that there is need to build long-term sustainable measures for the buffalo production system. Steps are needed to curb mortality of male buffalo calves in different parts of the country as these calves could otherwise be salvaged for providing quality meat for export and domestic consumption. As per the available data, 14 million male calves perish annually. If 70% can be salvaged, reared and processed @ 175 kgs per head—including edible offals – when reared they could yield 1.72 million tonnes of additional meat per annum.

Ministry of Agriculture, Department of Animal Husbandry, Dairy & Fisheries (DAHD&F), Government of India has announced an excellent scheme for Salvaging & Rearing of Male Buffalo Calves. However the scheme could not bring noticeable results and remained unviable particularly in the segment catering to "rearing upto 9 Buffalo calves by landless farmers". The Scheme has been redrafted for the 12th Plan but yet to be finally approved.

2. Traditional municipal slaughter houses face problems such as sanitation and hygiene, casual adherence to quality control and prevalence of manual slaughtering of animals. The Government of India is conscious of the quality issue in meat and is providing incentives to various state Governments in this regard. The Ministry of Food Processing Industries (MOFPI) has launched the scheme of 'Setting up of New/ Modernization of existing Abattoirs' with a view to ensure scientific and hygienic slaughtering of the animals and supply of quality meat and meat products. Under this scheme, grant-in-aid assistance of 50% in general areas and 75% in difficult areas, for cost of plant and machinery and technical civil work, upto ₹ 15 crore for each project is admissible.

The scheme has not been able to go beyond the establishment of a few slaughterhouses in metropolitan cities in the past. There is need for serious market study before initiation of the project as the scheme may not be feasible in all areas. Most large and medium sized projects cannot be run by local bodies on a sustainable basis independently after modernization. Some of the reasons cited for this are:

- a) High Operating Cost: The cost of operation of modern slaughterhouses is high due to high energy and fuel costs. The slaughtering fees collected at municipal slaughterhouses are insufficient to ensure working capital, which affects sustainability and economic feasibility.
 - b) The cost of modernizing the abattoir should be based on the (a) consumption trend of meat in that area, (b) demand-supply pattern, and (c) economic viability of the slaughter house and sustainability after implementation.
 - c) For rural and semi-urban areas, the low-cost model of modernising the abattoir should be adopted so that the same can run on sustainably, independently after its implementation.
 - d) Resistance among butchers to shift to new modern facilities, as they would incur higher transport costs and pay higher slaughtering fees.
 - e) There is need to implement such schemes at a wider scale with active participation from state Governments.
3. Less availability of trained manpower: Modernization of municipal slaughter house would require **trained manpower for respective operations**. Certified capacity building programme should be introduced for veterinarians, meat inspectors, butchers, etc. Training programmes could be conducted on various topics such as (a) butcher personal hygiene and care, (b) meat inspection, (c) meat stall management and environmental sanitation, (d) care and maintenance of butcher tools, (e) equipment hygiene, and (f) slaughtering and meat handling.

4. Prevalence of backyard slaughtering in the domestic value chain not only brings health hazards but also poses a threat to disposal of waste in a scientific manner. There is a dire need for dissemination and **implementation of clean technologies for slaughterhouse waste treatment**. This may include (a) proper facilities for skinning, (b) banning open disposal of viscera, and (c) proper collection of blood.
5. **Strengthening livestock markets:** Disparity has been found in the market and slaughtering fees in various states. In spite of collection of fees for transactions at livestock market which are held weekly, less effort is made by the local bodies or agricultural marketing committees to improve the conditions of the market. Keeping in view the immense benefit of such weekly markets, basic amenities for trouble-free marketing should be provided to livestock traders and farmers.
6. Norms for health certificate charges for export in line with Export of Raw Meat (Chilled / Frozen) (quality Control and Inspection) Rules, 1992, S.O 1989 dated 3rd September 1993 have been made and re-emphasized by Notification No. 82 of FTP Chapter 2 Note 8. It is recommended that all States should charge uniform health certificate charges in accordance with the established norms.
7. There is need for conscious focus on **utilisation of by-products** or value addition of by-products by municipal slaughter houses in comparison to export units. Waste products of slaughterhouses, like blood, rumen, and fat, etc. should be efficiently converted to fertilizer, tallow, MBM, and animal feed.
8. **Disease diagnostic activities and health care:** Animal diseases such as FMD require systematic planning for not only control but ultimate eradication. The FMD control programme started in only 54 districts in 2003 has now expanded to 221 districts. The availability of cold storage space for storage of FMD vaccines is very crucial. Realising the importance of such an aspect the private sector (meat export industry) is taking some timely initiatives by providing space for cold storage of FMD vaccines for vaccination programmes. They also provide transport for the state veterinarians to vaccinate. Such initiatives should be encouraged.
9. **Market Intelligence System:** Presently the numbers of animals reared by farmers in rural areas are small and they have limited marketable surpluses. However, with rising demand for buffalo meat in the domestic as well as international markets, farmers need education on aspects such as male buffalo calf rearing. Animal husbandry departments of State Government should develop market intelligence systems for creating reliable databases to assist in realistic planning.
10. **Availability of land:** Due to religious taboos, availability of land becomes difficult. Land being a vital component for establishment of new abattoirs becomes a major concern when (a) sufficient land is not available for construction, (b) there is difficulty in identification of suitable sites outside densely populated areas or away from residential areas, or (c) there exist complex norms for land conversion.

Recommendations For Export Oriented Buffalo Meat Value Chain

1. Availability of land: In addition to identification of suitable sites for construction of abattoirs, availability of sufficient land for construction of export oriented modern abattoirs which include building modern slaughter hall, lairage, ETP plants, etc. is also a major problem. Another two important related issues with land are (a) complex norms for conversion of land, (b) due to zero-discharge requirements of Pollution Control Boards larger land parcels are required for constructing abattoirs, which increases the overall cost.
2. Pollution control norms: Though pollution control norms are mandatory for a nation to create a safe environment, norms should be practical and easily adopted. According to the zero-discharge policy, even after maintaining the BOD (biological oxygen demand) level of less than 30mg/l and COD (chemical oxygen demand) of less than 250 mg/l, and TSS (100 mg/l), oil and grease (10 mg/l) meat packers are not allowed to discharge any effluent outside the export unit. On average 600-1000 litre water/day is used for buffalo in a slaughter house. Hence, it becomes difficult to utilise such a huge quantity of water inside the factory territory. Once the water is treated according to norms of CPCB it should be allowed to drain outside the factory territory.

Table 21. Characteristics of Slaughter House Waste Water

Particular	Concentration
pH	7.6-8.2
Total suspended solids (TSS), mg/L	2,500-3,000
Biochemical oxygen demand (BOD), mg/L	3,000-6000
Chemical oxygen demand (COD), mg/L	8,700-13,000
Oil and Grease, mg/L	200 -300
Waste water discharge, L/head	100-250*

Source: Central Pollution Control Board (CPCB)

Table 22. Waste Water Discharge Norms

	Parameters	Limit
Slaughter house	➤ 70 TL WK BOD	100 mg/l
	Suspended solids	100 mg/l
	Oil and grease	10 mg/l
Meat Processing	70 TL WK BOD	500 mg/l
	BOD	30 mg/l
	Suspended solids	50 mg/l
	Oil and grease	10 mg/l

Source: Central Pollution Control Board

TLWK- Tonnes of live weight killed (ii) In case of disposal into municipal sewer where sewage is treated, the industries shall install screen and oil & grease separation units. (iii) The industries having slaughter house along with meat processing units will be considered in meat processing category as far as standards are concerned.

3. For augmenting exports and assuring quality, developing a traceability model for Indian buffalo meat is very important. APEDA has laid down minimum requirements for construction of export oriented approved abattoirs. This requires identification of animals to be carried out as soon as the animal arrives at the slaughter house at the unloading area for slaughter. At the unloading area the animal has to be tagged by recording arrival time, sex and age along with source from which it has been drawn. This creates a preliminary traceability system, commencing as soon as the animal arrives. Thereafter traceability is enforced at various stages, namely, ante mortem, abattoir, post mortem, chilling and processing hall by assigning the batch number.
4. India is a member country of the World Organization for Animal Health (OIE) and is mandated to report list "A" and list "B" animal diseases to the OIE at regular intervals. Among the list "A" diseases of cattle and buffalo, except for foot and mouth disease, India is free from all other bovine diseases namely rinderpest, contagious bovine pleuropneumonia, lumpy skin disease, etc. However there is scope for implementing a country-wide disease control programme for FMD (foot and mouth disease) to tap the true potential of the export market. Department of Animal Husbandry, Dairying and Fisheries, Ministry of Agriculture and industry should come together to help the buffalo meat industry gain its a rightful position in target export markets by way of (a) taking advance preparatory actions, (b) maintaining availability of vaccines, (c) meeting OIE standards, (d) monitoring of disease, and (e) ensuring availability of cold stores for storing vaccines.
5. Shortage of feed and fodder is another area of concern in this sector. The decreasing area under fodder cultivation and reduced availability of crop residues as fodder should be given due attention.
6. Buffalo slaughtered in India are a by-product of the dairy industry and the average carcass yield per animal varies. APEDA considers a pan India average of 150 kgs. boneless meat and edible offals per buffalo. There should be efforts on the part of buffalo farmers to increase carcass yields.
7. Despite the immense potential of Indian buffalo meat in international market, it is not been able to cater to major international markets due to some occurrence of FMD, though at much reduced levels over the years. Vaccination programmes should be run by the government to control diseases like FMD and bovine contagious pleuropneumonia.
8. Proper utilisation of by-products and animal wastes are extremely vital and useful from the economic point of view. There are number of allied industries such as leather, poultry feed, handicraft, fish feed, and gelatin, which are by-products of the meat industry. Proper utilisation of animal by-products and animal wastes could be extremely vital and useful from the economic point of view. Technical knowhow to process animal by-products should be developed.
9. Transportation of animals: Animal welfare is an integral part of the livestock production system. Transportation of meat animals over long distances if not done properly and humanely results in loss of weight and can also add to unavoidable overhead costs. In such a situation, compliance with existing animal welfare laws should be ensured at every stage of the value chain. The meat export industry is increasingly involved in developing live buffalo transport systems in conformity with animal welfare norms.

10. State Governments should accord more priority to the livestock sector: Meat exports also help in raising the standard of living of small and marginal farmers who rear buffalo in their small backyards, and are able to sell their uneconomical buffalo to supplement their income. Various actors in the value chain, such as buffalo farmers, transporters, butchers, packing staff (generally females), and loading labourers are employed directly and indirectly in the livestock sector. Allied industries, such as leather, poultry feed, and handicrafts are dependent on the by-products of the meat industry.

For instance, Uttar Pradesh is emerging as a hub for export of buffalo meat for the country. UP has India's largest buffalo population (25% of total). Out of 9 agro climatic zones, 7 are favourable for buffalo production. From 2003 to 2007, 12.27% growth in buffalo population has been recorded. In UP, buffalo contribute more than 68% of milk production and 73% of meat production. There are approximately 645 cattle markets in the state. Likewise other states should also be encouraged to develop this sector.

11. Modernisation of municipal slaughterhouses which could be authorised for exports should be encouraged. To analyse the operation of one such model, where a municipal slaughterhouse is authorised for exports, a visit was made to the municipal slaughterhouse of Gazipur, Delhi.

Details about municipal abattoir, Ghazipur:

The "Delhi abattoir," as it is called, is situated near the border between New Delhi and Uttar Pradesh. This abattoir was established recently in place of the Idgah Slaughterhouse, which was abandoned as it was situated in the centre of the city and posed problems for pollution control and sanitation. For the first few years, it was run by the Municipal Corporation of Delhi (MCD) directly but was not viable. Hence, MCD sought to lease it to private entrepreneurs with some conditions, those being that three shifts have to be operated, two for the domestic market and one for export. The rendering plant and effluent treatment plant are also leased to Frigorifico Allana Limited (FAL)

Some observations:

1. Lease amount paid by Allanasons to the corporation is as follows:
 - a. ₹5,000,000 (USD 83,333) per month for the slaughterhouse,
 - b. ₹625,000 (USD 10,417) per month for the rendering plant.
2. The Delhi Municipal Corporation charges a fee of ₹300/head for slaughtering buffalo. Butchers are allowed to slaughter buffalo for sale of meat in the domestic market.
3. The production capacity of the slaughterhouse is as follows:
 - Buffalo – 1500 per day
 - Sheep and Goats
 - Halal – 4500 - 5000 per day
 - Non Halal – 250-350 per day
4. All costs for operation and maintenance are borne by Frigorifico Allana Limited and similarly revenues to their account.

5. There are fines for illegal slaughter within the city limits of Delhi:

- Buffalo – ₹2000 per animal
- Buffalo calf – ₹750 per calf

Policy Recommendations

11. The export duty on raw hides arising out of the operations of export oriented integrated buffalo abattoirs should be reconsidered.
12. “Tallow, fat and/or oils of any animal origin excluding fish oil” (Code no.1503) are prohibited/not permitted to be exported, as per the Export Policy Schedule-2. Permission for export of buffalo fat at least derived from buffalo livestock in integrated export oriented abattoirs needs to be considered.
13. There is need to strengthened the certification procedures. For instance, APEDA should issue shipment certificates to control the quantity each unit is producing to effectively implement the revised Foreign Trade Policy.

Direction for future Research

1. In view of inadequate information on post production aspects a study on meat and slaughter house by-products should be done. By-products of slaughter house can be classified as edible and inedible. A detailed study on (a) recovery and utilisation of edible by-products (b) production and utilisation of inedible by-products can be undertaken. Due to lack of necessary infrastructure facilities for collection and better utilisation of these by-products, optimum utilisation of these by-products has not been made and considerable quantities are wasted.
2. A study on innovative efforts made by industry should be documented such as (a) market upgradation and efficient handling, (b) Industry’s significant efforts in pollution control, and (c) efficient vehicles for proper transportation of animals.
3. Currently the meat is sold in fresh form immediately after slaughter in domestic market. A strong taboo exists among the population against consumption of chilled and frozen meat. Moreover, no surplus meat is being available for processing into various meat products. A study to understand the technologies used internationally for preparing value added products from buffalo meat should be done. Potential marketability of packed meat in the country should also be tested.
4. The modern abattoirs registered with APEDA are well equipped to cater to the demands of international market. However in view of various disadvantages and losses incurred in the traditional meat production and marketing system, a better alternative model of rural slaughterhouse which can undertake hygienic meat production and marketing, apart from processing of by-products should be developed for domestic market.



8

Glossary

Glossary

Abattoir	A slaughterhouse, or place where animals are sacrificed for food purposes.
Buffalo	Water buffalo of genus <i>Bubalus</i> spp.
Butcher	A grassroots level worker performing slaughter and dressing operations of food animals in slaughterhouses.
Carabeef	Meat of the water buffalo.
Cattle	Bovines of genus <i>Bos</i> spp.
Goshala	A sanctuary for cattle.
Lairage	A place where livestock are kept temporarily. A waiting, holding or recovery area supplied with appropriate feeding and watering. In American English, a “corral”, “holding pen”, or “feedlot.”
Offal	Internal organs and intestines of the animal. This term does not include muscle meat or bone. Offal can be further categorized as edible or inedible.
Rendering	The processing of waste animal tissue into stable, value-added materials, such as fats into usable lubricants, or waste bone into bone meal for poultry feed.
Slaughterhouse	The building, premises or place used for slaughtering of animals for human consumption.
Market fees/Mandi	Fees paid by purchasers to suppliers after the transaction for use of the market. Mandi fees vary from state to state in India. For instance, in Punjab a market fee of 1% of the sale price is charged.
Simple Random Sampling	A randomly selected sample from a larger sample or population, giving all the individuals in the sample an equal chance to be chosen. In a simple random sample, individuals are chosen at random and not more than once to prevent a bias that would negatively affect the validity of the result of the experiment.
Slaughtering fees	Fees charged by municipal corporations or other local bodies for slaughtering of animals. They are usually paid by the trader/butcher who owns the animal and slaughters it.
Snowball Sampling	One respondent is used to generate names of other potential respondents. It can be reiterated on a second set of respondents.
Stratified Random Sampling	A method of sampling that involves the division of a population into smaller groups (strata). In stratified random sampling, the strata are formed based on members' shared attributes or characteristics, such as age, income, sex, etc.

References

1. Census data 2001
2. Livestock Census 2003 & 2007
3. Agricultural and Processed Food Products Export Development Authority (APEDA)
4. Basic Animal Husbandry Statistics
5. Meat Processing Board
6. NABARD Consultancy Services (NABCONS)
7. FAO online database, 2011
8. Report of the National Commission on Cattle

Annexure 1:

Major conditions for disposal of pollutants from the buffalo industry

1. Standards to be achieved under water (prevention and control of pollution) act 1974.
2. Standards to be achieved under air (prevention and control of pollution) act 1974.
3. The industry shall not discharge any effluent (treated or untreated) onto land for stagnation/into any inland surface water at any time or any other unauthorized place by any unauthorized means and will discharge its treated effluent onto land for plantation only.
4. The industry shall not keep any outlet for discharge of outlet into any drain or any other inland surface water.
5. The industry shall not discharge any untreated effluent at any point of time.
6. The industry shall provide proper distribution system to discharge its treated effluent onto land for plantation, so that no stagnation occurs in the plantation area and the industry shall complete the plantation as well as distribution network in the 10 acre land by the end of February 2013 or before the commissioning of expansion project of the industry, whichever is earlier.
7. The industry shall provide adequate storage capacity for the treated effluent during no demand period as per CPCB guidelines for the proposed expansion.
8. The industry shall make proper disposal of the effluent so as to ensure that no stagnation occurs inside and outside the industrial premises and no demand period.
9. The industry shall ensure that the effluent discharged by it is toxicity free.
10. The industry shall install adequate and appropriate air pollution control device to control emissions from rendering plant.
11. The industry shall take all necessary steps to ensure that no odour/air pollution problem is caused in the area due to industrial operations after the proposed expansion.
12. The industry shall ensure that the emissions from the each stack shall conform to the emission standards laid down by the board from time to time.
13. The adequacy and efficacy of the effluent treatment plant/air pollution control devices will be the entire responsibility of the industry.
14. The industry shall get its emissions analysed from the board's lab/laboratory approved by the board at least four times in a year and shall submit the analysis results to the board.
15. The industry shall get its building plans approved under the provisions of section 3-A of Punjab factory rules, 1952.

16. The industry shall obtain cover under the public liability insurance act, 1991.
17. The industry shall obtain the NOC from the central ground water authority of India for the abstraction of ground water.
18. The industry shall ensure that no water/air pollution problem/public nuisance/odour problem is created in the area due to discharge of effluents/emissions from its industrial premises.

Environment Legislation

Pollution aspect	Legislation
Water Pollution	Water (Prevention and Control of Pollution) Act, 1974
Air and Noise pollution	Air (Prevention and Control of Pollution) Act, 1981
Solid waste	Municipal Solid Waste (Management and Handling) Rules, 2000

Annexure 2:

Overview of buffalo meat sector of nine states in India

1. ANDHRA PRADESH

Livestock Scenario

The livestock sector in Andhra Pradesh contributes around 5% of the state GSDP and 30% of its agricultural GSDP. The livestock sector recorded substantial growth in recent years, growing at an annual rate of 6.1% as against 6.4% in the crop sector during the period 1999-2000 to 2010-11. The growth rate of the livestock sector is more stable than that of the crop sector.

Animal Population

As per livestock census 2007, the state of Andhra Pradesh has around 9.3 million indigenous cattle, 1.9 million crossbred cattle, and 13.2 million buffalo. The population of crossbred cattle between 2003 and 2007 increased by 62%, the largest increase being in coastal Andhra Pradesh. Surprisingly, the inventory of indigenous cattle also increased by 19%, while it had shown a decline of 11% during 1999-2003. The buffalo inventory increased by 24% during 2003-07, the largest increase being in the Telengana region. The livestock statistics are reproduced in the table below:

Andhra Pradesh	Livestock Census (million)		% Change in 2007 over 2003
	2003	2007	
Cattle	9.3	11.2	(+) 20
Buffalo	10.6	13.2	(+) 24

Source: Livestock Census, GoI

Buffalo Slaughtered & Carabeef Production

- Buffalo Slaughtered (in '000) in 2011-12: 1,082
- Avg. Carcass Yield/ Animal (Kg) in 2011-12: 103
- Carabeef Production ('000 tons) in 2011-12: 112

The number of buffalo slaughtered per annum is increasing gradually. Number of buffaloes slaughtered and the carabeef production have doubled over the last four years. Carabeef production in 2011-12 was 112 thousand tons, an increase of 58% over 2007-08. This indicates that demand for carabeef has been rising constantly during the last few years.

Population (by religion)

The Muslim population constitutes around 9.17% of the total population of Andhra Pradesh. Muslims are the major carabeef-consuming segment population in Andhra Pradesh. Hindus constitute 89.01% and Christians 1.55% of the total population of Andhra Pradesh.

Major Municipal Slaughterhouse

There are 196 Government owned and 273 unorganized slaughterhouses. Two private slaughterhouses have state-of-the-art slaughter and processing facilities. In addition, there are two carcass utilization centers. Municipal Slaughterhouses: Chengicherla, Ramnaspura and Amberpet.

Major Export Units: Approved Indian Abattoirs-Cum Meat Processing Plants

1. Al Kabeer Exports (P) Ltd.
2. Frigerio Conserva Allana Ltd.
3. Chengicherla Slaughterhouse

Regulations

Slaughter of cows (genus *Bos* spp.) of either sex and of any age, or of buffalo calves is prohibited. No certificate shall be issued if in the opinion of the competent authority the animal is or is likely to become economical for the purpose of:

- a) Breeding, or
- b) Draught or any kind of agricultural operations, or
- c) Giving milk or bearing offspring.

2. BIHAR

Livestock Scenario

Animal husbandry is one of key livelihoods and employment opportunities for the rural masses, and contributes about one-fifth of total rural income. This sector also generates large-scale employment to women belonging to the marginalized strata of society. The contribution of agriculture to total GDP of the state was 33.1% and that of the livestock sector to agricultural GDP was 38.7% at current prices for 2005-06.

Animal Population

According to the 2007 livestock census, the total livestock inventory of Bihar is 41.613 million with a cow herd of 12.4 million and a buffalo herd of 6.7 million. The indigenous-breed cow population has been declining, while the crossbred population has been increasing rapidly and the buffalo population has been growing marginally.

Bihar	Livestock Census (million)		% Change in 2007 over 2003
	2003	2007	
Cattle	10.7	12.4	16
Buffalo	5.7	6.7	18

Source: Livestock Census, GoI

Buffalo Slaughtered & Carabeef Production

- Buffalo Slaughtered (in 000) in 2011-12: 693
- Avg. Carcass Yield/ Animal (Kg) in 2011-12: 62
- Carabeef Production (000 tons) in 2011-12: 43

Carabeef produced in 2011-12 was 43 thousand tonnes, up 2% from the 2010-11 figure of 42 thousand tonnes. Carabeef production in the state has been increasing very steadily since 2007-08. The number of buffaloes slaughtered also rose by 2% in 2011-12 over the previous year.

Population (by religion)

In Bihar, Hindus form the major part of the population with 84% followed by Muslims with 16.6%. Christian numbers are negligible. The Muslim community is the major consumer of carabeef in the state.

Major Municipal Slaughterhouses

There are 42 registered slaughterhouses in Bihar, as follows:

Total Registered Slaughterhouses - 42.

S.no.	District	Registered Slaughter-houses
1.	Patna	1
2.	Bhojpur	1
3.	Rohtas	1
4.	Gaya	4
5.	Aurangabad	1
6.	Navada	1
7.	Muzaffarpur	1
8.	Vaishali	2
9.	West champaran	3
10.	East Champaran	4
11.	Sitamarhi	1
12.	Chappra	1
13.	Siwan	1
14.	Gopalganj	1
15.	Darbhanga	3
16.	Madhubani	2
17.	Samastipur	1
18.	Bhagalpur	2
19.	Munger	3
20.	Khagariya	1
21.	Purnia	4
22.	Kishanganj	2
23.	Arariya	1
24.	Katihar	1
Total		42

Source: Department of Animal Husbandry, Bihar

Regulations

List of laws enacted by states for cow protection: The Bihar Preservation and Improvement of Animals Act, 1955.

3. KARNATAKA

Livestock Scenario

The animal husbandry and dairy sectors in Karnataka contribute 2.97% of the state's GDP and 22% of the agricultural GDP.

Animal Population

The cattle and buffalo herds have grown in recent years. Total breedable cows number 4.913 million and buffalo 2.881 million. The aggregate herd of cows grew 10% and of buffalo 8.2% between 2003 and 2007. Karnataka's share of the national bovine herd is 5.83%.

Karnataka	Livestock Census (million)		% Change in 2007 over 2003
	2003	2007	
Cattle	9.53	10.5	10
Buffalo	3.99	4.3	8

Source: Livestock Census, GoI

Buffalo Slaughtered & Carabeef Production

- Buffalo Slaughtered (in '000) in 2011-12: 90
- Avg. Carcass Yield/ Animal (Kg) in 2011-12: 107
- Carabeef Production ('000 tons) in 2011-12: 10

The slaughter number of buffaloes has increased very slowly, up 7% in the last four years, though buffalo meat production was up 10% over the same period. Carabeef production in 2011-12 was 10,000 tons.

Population (by religion)

83% of the population is Hindu, 12.2% is Muslim, and 3.1% is Christian. Muslims are the major segment of the population consuming carabeef. Buffalo meat is not a preferred meat for the people of Karnataka but in recent times there is a shift among lower-income groups from mutton (goat meat) to buffalo meat because of the rising price of mutton.

Major Municipal Slaughterhouse

Total number of slaughterhouses in the state is 96, listed below. The Civil Meat Production & Processing Centre (MPPC) is the municipal slaughterhouse in Bangalore.

District wise Slaughterhouses as of 2011-12:

S.no.	District	Registered Slaughterhouses
1	Bangalore (U)	5
2	Bangalore (R)	1
3	Belgaum	2
4	Bellary	4
5	Bidar	11
6	Bijapur	14
7	Bagalkote	2
8	Chickmangalur	8
9	Chitradurga	1
10	Chickballapur	0
11	D. Kannada	1
12	Dharwad	2
13	Gulbarga	2
14	Hassan	0
15	Kodagu	6
16	Kolar	6
17	Mandya	1
18	Mysore	1
19	Raichur	2
20	Shimoga	2
21	Tumkur	1
22	U. Kannada	1
23	Ramnagar	3
24	Chamarajanagar	0
25	Devanagere	2
26	Gadag	2
27	Haveri	16
28	Koppal	0
29	Udupi	0
30	Yadgir	0
	State Total	96

Source: Department of Animal Husbandry, Karnataka

Regulations

The Karnataka Prevention of Cow Slaughter and Cattle Preservation Act, 1964, as amended in 1975, prohibits the slaughter of cows or of calves of she buffalo. A certificate for slaughter shall be granted if a) the buffalo is over the age of twelve years, or b) the buffalo has become permanently incapacitated for breeding, draught or giving milk due to injury, deformity or any other cause.

4. KERALA

Livestock Scenario

About 10% of the gross domestic product of Kerala is contributed by this sector. The cattle and buffalo herds have fallen in recent years. Slaughter of cattle (genus *Bos* spp.) is legal in Kerala.

Animal Population

As per the livestock census 2007, the cattle herd in the state numbers 1,740,117 head, which indicates a decrease of 18.01% from the 17th census. The 18th livestock census shows a decrease in crossbred cattle of 6.57% and of indigenous cattle by 69.29% between 2003 and 2007.

The buffalo inventory in the 18th Livestock census was 58,145 head. The buffalo population has decreased by 10.01% between 2003 and 2007.

Kerala	Livestock Census (million)		% Change in 2007 over 2003
	2003	2007	
Cattle	2.1	1.7	-18
Buffalo	.06	.05	-10

Source: Livestock Census, GoI

Buffalo Slaughtered & Carabeef Production

- Buffalo Slaughtered (in '000) in 2011-12: 896
- Avg. Carcass Yield/Animal (Kg) in 2011-12: 102
- Carabeef Production ('000 tons) in 2011-12: 91

The number of buffaloes slaughtered per annum is increasing gradually. Buffalo slaughter and meat production almost doubled during the last four years. Buffalo meat production in 2011-12 was 9.1 million tons. Buffalo meat production increased by 98% since 2007-08. This indicates that demand for carabeef has doubled during the last few years.

Population (by religion)

Hindus constitute 56.2% of the total population of Kerala, Muslims 24.3%, and Christians 19%. The Muslim population in Kerala is 5.6% of the total Muslim population of India. Muslims and Christians together are the major segment of the population consuming carabeef in Kerala.

District wise Slaughterhouses in Kerala: 2011-12

S.No.	District	Panchayath/Municipality/ Corporation	No.
1	Thiruvananthpuram	Karavaram	6
		Parassala	2
		TVM Corporation	3
		Cherunniyoor	1
		Vettor	1
		Attingal Municipality	1
		Pazhayakunnummel	1
		Kallara	1
		Neyyattinkara Municipality	2
		Nedumangad Municipality	1
		Total	19
2	Kollam	Kollam Corporation	1
		Paravoor Municipality	1
		Punalur	1
		Total	3
3	Panthanamthitta	Pandalam	3
		Anicadu	1
		Omallur	3
		Thiruvalla	1
		Total	8
4	Alappuzha	Kayamkulam Municipality	1
		Total	1
5	Kottayam	Kottayam Municipality	1
		Vaikom	1
		Paippad	1
		Vazhavor	1
		Erattupetta	1
		Thalayolapparambu	1
		Total	6

S.No.	District	Panchayath/Municipality/ Corporation	No.
6	Idukki	Adimaly	1
		Elappara	1
		Kanjikkuzhi	1
		Karunapuram	1
		Kattappana	4
		Kumily	12
		Nedumkandam	1
		Peerumedu	2
		Rajakkad	1
		Vannapuram	1
		Vathikkudy	1
		Total	26
Total			89

Source: Department of Animal Husbandry, Kerala

Regulations

Slaughter of cows as well as buffalo is permitted in Kerala. Beef is widely consumed in Kerala and is the cheapest meat available. As slaughter of cows is banned in neighbouring Karnataka and Tamil Nadu, large numbers of spent dairy cows and buffalo are illegally smuggled into the state. According to the state economic review for 2009, there are no reliable data on numbers of cattle being slaughtered in Kerala. In a survey conducted in 2006, the Government had identified 904 slaughterhouses, of which only 30 were authorized.

5. MAHARASHTRA

Livestock Scenario

Livestock play an important role of providing supplementary income opportunities to rural households having agriculture as their main livelihood. The contribution of the livestock sector to gross state agricultural product during 2006-07 was about 25%. The state's share of India's livestock herd is 6.8 per cent. This state has the sixth highest buffalo inventory in the country. Maharashtra permits slaughter of unproductive bullocks.

Animal Population

As per the 17th livestock census, Maharashtra had 8.8% of the cattle and 6.28% of the buffalo in the country. In Maharashtra, numbers of crossbred cattle have increased by 12.9% but indigenous cattle decreased by 13.4% during the period between 2003 and 2007. The growth rate was minimal in case of indigenous cattle, while crossbred cattle and buffalo ranged from 4% to 5% growth between 2003 and 2007.

The cattle herds decreased, by 3.3%, between 2003 and 2007. And the overall proportion of buffalo in the state has been increasing at the rate of 0.75 per cent per annum during the most recent 10 years of data (1997-2007).

Maharashtra	Livestock Census (million)		% Change in 2007 over 2003
	2003	2007	
Cattle	16.7	16.2	(-)3
Buffalo	6.1	6.1	0

Source: Livestock Census, GoI

Buffalo Slaughtered & Carabeef Production

- Buffalo and Bullocks Slaughtered (in '000) in 2011-12: 658
- Avg. Carcass Yield/ Animal (Kg) in 2011-12: 139
- Carabeef Production ('000 tons) in 2011-12: 92

Of the total estimated meat production, the contribution from cattle (genus *Bos* spp.) is about 12.05%, whereas from buffalo it is 18.20%. The number of animals slaughtered (Buffalo and Bullocks) is increasing annually but there was a decline of 8% in 2012. Carabeef production in the state had been increasing at the rate of 1-2% annually since 2007-08 but in 2011-12 it decreased by 7%. Slaughter numbers and carabeef production have been declining during 2011-12, resulting in reduced availability of carabeef in Maharashtra.

Population (by religion)

In Maharashtra, Hindus forms the religious majority with 80.2%, followed by Muslims with 10.6%. Christians constitute 1%. The Muslim community is the major consumer of carabeef in the state.

Major Municipal Slaughterhouses

There are around 338 registered slaughterhouses in the state (see list below). Deonar Abattoir, located in Deonar, an eastern suburb of Mumbai, is the municipal slaughterhouse in Mumbai.

S.No.	District	No. of registered Slaughterhouses
1	Brihan Mumbai	1
2	Tahne	3
3	Raigad	1
4	Sindhudurg	2
5	Nashik	10
6	Dhule	4
7	Nandurbar	5
8	Jalgaon	19
9	Ahmednagar	7
10	Pune	8
11	Satara	12
12	Sangali	2
13	Kolhapur	13
14	Solapur	4
15	Latur	6
16	Osmanabad	4
17	Beed	11
18	Aurangabad	11
19	Jalna	5
20	Buldhana	47
21	Parbhani	6
22	Nanded	17
23	Hingoli	11
24	Akola	28

S.No.	District	No. of registered Slaughterhouses
25	Washim	14
26	Amravati	26
27	Yavatmal	30
28	Vardha	6
29	Nagpur	14
30	Bhandara	1
31	Gadchiroli	3
32	Chandrapur	7
	State Total	338

Source: Department of Animal Husbandry, Maharashtra

Major Export Units: Approved Indian Abattoirs-Cum Meat Processing Plants

1. Frigorifico Allana Ltd.
2. Deonar Abattoir
3. Son Anku Exports (P) Ltd

Regulations

The Maharashtra government permits slaughter and production of beef through authorized abattoirs set up by local municipal corporations. Bullocks and buffalo above the age of 12-13 years may be slaughtered. Officials say only animals which cannot reproduce or are of no use for agricultural purposes are allowed to be slaughtered. Maharashtra state government is yet to align the Health Certification charges to those specified under the Export of Raw Meat (Chilled/ Frozen) (Quality Control and Inspection) Rules, 1992, S.O 1989 dated 3rd Sept 1993 and emphasized in Chapter 2 of FTP, Note 8, Notification No. 82 dated 31st October 2011.

6. PUNJAB

Livestock Scenario

Livestock contributed 7.25% to the gross state domestic product (GSDP) at constant (2004-05) prices in 2011-12. The livestock sector, which was previously enjoying an upswing, has been decelerating since 1997. As per the last 17th livestock census, the state of Punjab had 1.10% of the cattle and 6.12% of the buffalo in India.

Animal Population

The overall livestock population in Punjab has been decreasing continuously since 1990. There was a tremendous decline from about 9.7 million head in 1990 to only 7.1 million head by 2007, a fall of 1.5 per cent per annum. This included a drastic reduction in crossbred cattle as well as buffalo between 2003 and 2007.

The cattle herd in Punjab has declined from about 2.8 million head in 1990 to about 1.8 million as of 2007, a decrease of 2.29 per cent per annum. The buffalo inventory in Punjab has declined from about 5.6 million head in 1990 to 5.0 million as of 2007, falling at the rate of 0.69 per cent per annum.

Punjab	Livestock Census (million)		% Change in 2007 over 2003
	2003	2007	
Cattle	2.0	1.8	-10
Buffalo	6.0	5.0	-17

Source: Livestock Census, GoI

Buffalo Slaughter & Carabeef Production

- Buffalo Slaughtered (in '000) in 2011-12: 736
- Avg. Carcass Yield/Animal (Kg) in 2011-12: 27
- Carabeef Production ('000 tons) in 2011-12: 93

The number of buffaloes slaughtered per annum is increasing gradually. Buffaloes slaughtered and the carabeef production have been more than doubled during the last four years for which data were available. Meat production in 2011-12 was 93,000 tons. Buffalo meat production has increased by 50% since 2007-08. This indicates that demand for carabeef has been rising constantly during the last few years.

Population (by religion)

In Punjab, Sikhism is the main religion with 60% of the population, followed by Hindus with 37%. Muslims and Christians constitute a small part of the population of around 1 to 2%. There is no local consumption of carabeef in Punjab; all carabeef produced is being exported to other countries.

Major Municipal Slaughterhouses

Municipal slaughterhouses for buffalo do not exist in Punjab. There are four private slaughterhouses, which produce solely for export.

Major Export Units: Approved Indian Abattoirs- Cum Meat Processing Plants

1. M K Overseas (P) Ltd.
2. Abbott Cold Storages Pvt. Ltd
3. Mirha Exports Pvt. Ltd
4. PML Industries

Regulations

Cow (genus *Bos* spp.) slaughter is prohibited in Punjab. Under the Punjab Prohibition of Cow Slaughter Act, 1955, whoever contravenes or attempts to contravene this law is punishable under rigorous imprisonment for a term of up to two years or with a fine of up to ₹ 1000, or both. A proposal to set up a Cow Service Board is under consideration by the state Government.

7. RAJASTHAN

Livestock Scenario

Livestock contribute approximately 9% to the state's GDP.

Animal Population

The cattle and the buffalo herds are growing substantially in the state. There was an increment of 11% in cattle and 6% in buffalo in 2007 over 2003.

Rajasthan	Livestock Census (million)		% Change in 2007 over 2003
	2003	2007	
Cattle	10.9	12.1	11
Buffalo	10.4	11.1	6

Source: Livestock Census, GoI

Buffalo Slaughtered & Carabeef Production

- Buffalo Slaughtered (in '000) in 2011-12: 324
- Avg. Carcass Yield/ Animal (Kg) in 2011-12: 70
- Carabeef Production (000 tons) in 2011-12: 23

Buffalo meat produced in 2011-12 totaled 23 thousand tonnes, which remained unchanged as compared to the prior year. Buffalo meat production in the state has almost doubled since 2007-08. Both the slaughter numbers and carabeef production had been increasing but levelled off between 2010-11 and 2011-12.

Population (by religion)

Rajasthan's population is made up mainly of Hindus, who account for 88.8% of the population. Muslims make up 8.5%, Sikhs 1.4% and Jains 1.2% of the population. The state of Rajasthan is also populated by Sindhis, who came to Rajasthan from Sindh province (now in Pakistan).

Major Municipal Slaughterhouses

1. Chainpura Slaughterhouse, Jaipur
2. Municipal Slaughterhouse, Tonk
3. Only buffalo are slaughtered in these slaughterhouses.
4. Export oriented slaughter house : Prime Natural Food Pvt Ltd.
5. Fragile Frozen Food

Regulations

Slaughter of cows is prohibited in the state under the Rajasthan Bovine Animal (Prohibition of slaughter and regulation of temporary migration or exports) Act, 1995. Possession for sale, sale, and transport of beef and beef products are prohibited. Export of bovine animals for slaughter is prohibited. Custody of seized animals is to be given to any recognized voluntary animal welfare agency, or to a goshala or gosadan. Violation of the above invites imprisonment of one to two years and a fine of up to ₹10,000.

(Bovine literally includes cow and buffalo. But in Rajasthan buffalo slaughter is permitted whereas cow slaughter is banned).

8. UTTAR PRADESH

Livestock Scenario

Uttar Pradesh is one of the largest states in the country with a large livestock population, representing 10% of the cattle and 23% of the buffalo in India. Uttar Pradesh ranks third in meat production and of all states is the biggest exporter of buffalo meat. Out of total Indian carabeef exports, 67% originates in Uttar Pradesh.

Animal Population

As per the 18th Livestock Census of 2007, there are 19 million head of cattle and 26.4 million buffalo. The buffalo herd is growing substantially in this state. There is an increment of 3% in cattle and 15.38% in buffalo in 2007 over 2003.

As per the 18th livestock census, in 2007 the state had 10.02% of India's cattle and 23.4% of its buffalo. This equates to the second highest cattle inventory and highest buffalo inventory in the country. In Uttar Pradesh, crossbred cattle decreased by 22.4% and indigenous cattle decreased by 5.6% during the period between the 16th and 17th census, for an overall decrease of 7.3% in the total cattle herd during the inter-censal period. The buffalo population, however, increased sharply.

Uttar Pradesh	Livestock Census (million)		% Change in 2007 over 2003
	2003	2007	
Cattle	18.5	19.0	3
Buffalo	22.9	26.4	15

Source: Livestock Census, GoI

Buffalo Slaughtered & Carabeef Production

- Buffalo Slaughtered (in '000) in 2011-12: 3242
- Avg. Carcass Yield/ Animal (Kg) in 2011-12: 126
- Carabeef Production ('000 tons) in 2011-12: 408

Buffalo meat produced in 2011-12 was 408 thousand tonnes, up 36% from 2010-11 output of 300 thousand tonnes. Buffalo meat production in the state has been increasing gradually since 2007-08 but since 2010 it increased by 40%. Buffalo slaughter numbers also increased by 40% in 2011-12 from the prior year.

Population (by religion)

In Uttar Pradesh, Hindus form the majority with 80.6%, followed by Muslims with 18.4%. U.P. is the state with highest Muslims population in the country. Numbers of Christians are negligible. The Muslim community is the major consumer of beef in the state.

Major Municipal Slaughterhouses

There are 317 registered slaughterhouses in the state and in addition to this there are 24 export-oriented units for buffalo meat. The major municipal slaughterhouses in Uttar Pradesh are Aishbagh and Fatehganj.

Major Export Units: Approved Indian Abattoirs-Cum Meat Processing Plants

- | | |
|--|--|
| 1. Hind Agro Industries | 12. Eagle Continental Foods Pvt. Ltd |
| 2. Indagro Foods Ltd. | 13. Al Hamd Frozen Foods (P) Ltd |
| 3. Amroon Foods Pvt. Ltd. | 14. Al Faheem Meatex (P) Ltd. |
| 4. Al Saquib Exports Pvt. Ltd | 15. HMA Agro Industries Ltd. |
| 5. Medina Frozen Food Exports Pvt. Ltd | 16. Frigerico Conserva Allana Ltd. |
| 6. Al Noor Exports | 17. AOV Exports Pvt. Ltd. |
| 7. Al Hamd Agro Food Products (P) Ltd | 18. Rustam Foods Pvt. Ltd. |
| 8. J.S International | 19. Nagar Nigam Slaughterhouse, Saharanpur |
| 9. Fair Exports India (P) Ltd. | 20. Rayban Foods Private Ltd. |
| 10. Al Nafees Frozen Food Exports Pvt. Ltd | |
| 11. MCD Slaughterhouse | |

Since the business of buffalo meat processing is well established and accepted in the State of Uttar Pradesh, many units are operating which are procuring buffalo not only from UP, but neighbouring States like Bihar, Rajasthan, Chattisgarh, Jharkhand, Madhya Pradesh, Haryana and Punjab also.

Regulations

Cow (genus *Bos* spp.) slaughter has been banned in Uttar Pradesh since 1955. The Uttar Pradesh Cow Slaughter (Prevention) Act, 1955, banning cow slaughter, was enacted in 1955. However the law permitted slaughter of old and dry cows after receipt of prior permission from the animal husbandry department. In 2002 the Government amended the act, imposing a blanket ban on cow slaughter. As per the Amendment Act 2001, no person shall slaughter or cause to be slaughtered or offer or cause to be offered for slaughter a cow, bull or bullock in any place in Uttar Pradesh, anything contained in any other law for the time being force or any usage or custom, to the contrary notwithstanding. Any person may surrender his cow, bull or bullock to a goshala or an institution which shall accept such cow, bull or bullock according to the availability of accommodation. Any cow, bull or bullock so surrendered shall not be returned to such person.

9. WEST BENGAL

Livestock Scenario

The animal husbandry and dairy sectors in West Bengal are currently growing at 3% per annum. Development of the animal husbandry and dairy sectors in West Bengal state has been essentially a case of ample opportunities. These sectors contribute 5% to state domestic product (SDP) and nearly 20% of agricultural production. Slaughter of both cattle and buffalo is permitted, but only if the animal is over 14 years of age and is unfit for work or breeding, or has become permanently incapacitated for work and breeding due to age, injury, deformity, or any incurable disease.

Animal Population

As per the 2007 livestock census, the state had 16.54 million indigenous cattle, 2.64 million crossbred cattle, and 0.76 million buffalo. Inventories of both indigenous and crossbred cattle rose, while buffalo numbers fell.

West Bengal	Livestock Census (million)		% Change in 2007 over 2003
	2003	2007	
Cattle	18.0	19.2	7
Buffalo	0.84	0.76	-10

Source: Livestock Census, GoI

Buffalo Slaughtered & Carabeef Production

- Buffalo Slaughtered (in '000) in 2011-12: 116
- Avg. Carcass Yield/ Animal (Kg) in 2011-12: 144
- Carabeef Production ('000 tons) in 2011-12: 17

Buffalo slaughter numbers per annum are increasing, with both slaughter and buffalo meat production up by more than 50% over the last four years for which data is available. Buffalo meat production in 2011-12 was 17 thousand tons, up 70% since 2007-08. This indicates that demand for carabeef has been rising during the last few years.

Population (by religion)

Muslims constitute 25.2% of the population of West Bengal and are the major segment of the population which consumes carabeef. Hindus are 72.5% and Christians 0.64% of the total.

Major Municipal Slaughterhouse

There are two municipal slaughterhouses in West Bengal.

A. Tangra Slaughterhouse: On average 450 bovines/day are halal slaughtered in batches

B. Slaughterhouse at Durgapur

Regulations

Under the West Bengal Slaughter Act , 1950, one needs to produce a 'fit-for-slaughter' certificate issued jointly by the president of a municipality or a panchayat samiti and a veterinarian. The animals included in the Act are bulls, bullocks, cows, calves and buffalo of all types/ages. The certificate is issued if the animal is over 14 years of age and is unfit for work or breeding, or has become permanently incapacitated for work and breeding due to age, injury, deformity, or any incurable disease.

Annexure 3:

Regulations and Laws across various states with regard to cow/buffalo slaughter

State	Law	Details
Delhi	Delhi Agricultural Cattle Preservation Act, 1994	Cow slaughter is banned. The law imposes a total prohibition on the slaughter of cows, calves, bulls and bullocks defined as "agriculture cattle".
Uttar Pradesh	The Uttar Pradesh cow slaughter (prevention) Act, 1955 Amendment in 2002	Blanket ban on cow slaughter, no person shall slaughter or cause to be slaughtered or offer or cause to be offered for slaughter a cow, bull or bullock in any place in Uttar Pradesh.
Rajasthan	Rajasthan bovine animal Act , 1995	Slaughter of "bovines" is prohibited Section 2(b) of the Act defines a "Bovine animal" to mean and include Cow, Calf, Heifer, Bull or Bullock. Section 2(e) defines "Calf" to mean a castrated or uncastrated male of the age of three years and below belonging to the species of bovine animal. (Bovine literally includes cow and buffalo. But in Rajasthan buffalo slaughter is permitted whereas cow slaughter is banned).
Jharkhand	There is no law related to beef consumption	Cow slaughter is permitted.
Tamil Nadu	Tamil Nadu animal prevention Act, 1958	Slaughter of cows and heifers is banned in all slaughterhouses in Tamil Nadu.
Karnataka	Prevention of cow slaughter and cattle preservation Act, 1964. Amended in 1975	Prohibits the slaughter of cow or calf of she buffalo. A certificate for slaughter shall be granted for (a) the animal is over the age of twelve years or (b) the animal has become permanently incapacitated for breeding, draught or giving milk due to injury, deformity or any other cause.
Odisha	Prevention of cow slaughter Act, 1960	Killing of cows totally prohibited. Bull, bullock is allowed on production of fit for slaughter certificate if the animal is over 14 years of age or has become permanently unfit for breeding.

State	Law	Details
Gujarat	The Bombay animal preservation Act, 1954 (applied to gujarat),	No certificate for slaughter shall be granted in respect of a cow, the calf of a cow, whether male or female and if male, whether castrated or not, a bull or a bullock; total ban on slaughter of bull and bullock was not upheld by the honourable Gujarat high court.
Madhya Pradesh	Gauvansh Vadh Pratishedh (Sanshodhan) Vidheyak, 2007	Total ban on cow slaughter.
Andhra Pradesh	The Andhra Pradesh Prohibition Of Cow Slaughter And Animal Preservation Act, 1977	Slaughter of cows(includes a heifer or a calf, male or female of a cow) or calves of she buffaloes is prohibited No certificate shall be granted to slaughter if the animal is fit for • Breeding, • Draught or any kind of agricultural operations, • Giving milk or bearing offspring, • Buffalos below age of 8 are not allowed.
Kerala	No state legislation - only Panchayat Act / Rules	Beef is widely consumed in Kerala and is the cheapest meat available.
Assam	Assam cattle preservation Act, 1951, Amended in 1963	Prohibits slaughter of cattle. No certificate shall be issued unless • The cattle is over 14 yrs, • Unfit for working o breeding, • The animal has become permanently incapacitated.
Punjab	Cow slaughter Act, 1955	Complete ban on cow slaughter in Punjab.
Himachal Pradesh	Cow slaughter Act. 1979	Cow slaughter is banned.
Maharashtra	The Maharashtra Animal Preservation Act, 1976	Permits slaughtering and production of beef through authorized abattoirs set up by local municipal corporations. Buffalo and bullocks above the age of 12-13 years can be slaughtered.
Jammu and Kashmir	The Ranbir Penal Court, 1932	Cow slaughter is banned. Buffalo allowed for slaughter and industrial production of meat for domestic and export of meat.

State	Law	Details
West Bengal	The West Bengal Animal Slaughter Act, 1950	Fit for slaughter certificate is required. Animals included in the Act are bulls, bullocks, cows, calves and buffaloes of all types/ages. Animal must be over 14 years of age and unfit for work or breeding, or have become permanently incapacitated for work and breeding due to age, injury, deformity, or any incurable disease.
Goa	The Goa, Daman & Diu prevention of cow slaughter Act, 1978. And The Goa animal preservation Act, 1995 Goa animal preservation Act , 1995, Amended in 2010.	No person shall slaughter or cause to be slaughtered any scheduled animal in any place in the state of Goa, unless he has obtained in respect of such animal a certificate in writing from the competent authority that the animal is fit for slaughter No certificate shall be provided if in the opinion of the competent authority • The scheduled animal, if male is likely to become economical for the purpose of draught or any kind of agricultural operation, • For breeding, if male, • If female, milking, bearing offspring.

Source: Meat & Poultry Processing Board, Ministry of Food Processing Industries, GOI, report on “ Benchmarking of Abattoirs” prepared by NABARD Consultancy Services (NABCONS), 2013

Annexure 4:

State-wise information on slaughter houses

S.N.	State/ UT	No. of slaughter houses	No. of animals slaughtered/day		
			Small	Large	Total
1	Andhra Pradesh	130	15623	4321	19944
4	Bihar	3			
5	Chhattisgarh	11	236		236
6	Goa	1		150	150
7	Gujarat	39	936	144	1080
8	Haryana	21	445		445
9	Himachal Pradesh	25	318		318
10	Jammu & Kashmir	4	300		300
12	Karnataka	99	4929	679	5608
13	Kerala	47	666	599	1265
14	Madhya Pradesh	25	712	263	975
15	Maharashtra	150	8558	9285	17843
17	Meghalaya	4	105	10	115
18	Mizoram	1	2	3	5
21	Punjab	45	2655	1710	4365
22	Rajasthan	7	2690	100	2790
23	Sikkim	0		12	
24	Tamil Nadu	111	6527	405	6932
26	Uttar Pradesh	126	7641	10519	18160
27	Uttaranchal	10	347	27	374
28	West Bengal	42	2898	332	3230
30	Chandigarh	1	200		200
31	Daman Diu Dadra Nagar Haveli	2	20	4	24
32	Delhi	1	2000	500	2500
33	Lakshadweep	1		1	1
34	Puducherry	1	70	5	75
	Total	907	57878	29057	86935

Source: Presentation by a Senior Official from Central Pollution Control Board, given at the 6th Mayor Conference on September 13, 2013 at New Delhi.

Annexure 5:

Traditional communities engaged in slaughter of meat animals and sale of meat

Muslims are the predominant group engaged in the meat industry, but some other communities are also engaged in meat animal trade and the meat business. The traditional Muslim community involved in slaughtering is referred to as “qassab”, the Arabic word for butcher. The qassabs work in abattoirs as well as meat shops, where they process and sell meat of buffalo, poultry, goats, and sheep, as well as seafood and fish. The most common surname of the qassab community is Qureshi but not all Qureshis are qassabs. They are mainly found in Uttar Pradesh, Bihar, Rajasthan, Maharashtra and Jammu and Kashmir.

Traditional Communities engaged in slaughter of meat animals and sale of meat

S. no.	Community	Origin	Religion	Animal Slaughtered	Distribution
1	Khatik	Kshatriya	Both Hindu & Muslim	Swine and Goats	Throughout India but mostly in Punjab, Haryana, Rajasthan, Gujarat, Delhi, Uttar Pradesh, Madhya Pradesh, Andhra Pradesh, Maharashtra, and Bihar.
2	Qassab (also known as Chikawa, Bare Qasai, Baker Qasai, Kasai)	Arabs of the Qureshi tribe	Muslim	Poultry, Seafood, Cattle, Goats, Sheep, etc.	North India, Pakistan, Uttar Pradesh, Bihar, Rajasthan, Maharashtra, Jammu and Kashmir
3	Chik (a subgroup of Qassab, also known as Bakar Qasab, Buz Qasab, Chikwa)	Arabs of the Qureshi tribe	Muslim	Cows and Buffalo	Uttar Pradesh, Bihar
4	Bhadik	N.A.	Hindu		Uttar Pradesh

Source: Meat & Poultry Processing Board, Ministry of Food Processing Industries, GOI, report on “Socio Economic Upliftment of Meat Industry Workers” prepared by NABARD Consultancy Services (NABCONS), 2013

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